

ELOG Field Analyzer

Ver. 1.1.0

User's Manual

2026 / 1 / 31

NT System Design

Table of Contents

1. Introduction.....	1
2. License	1
3. Operating Environment	6
4. Installation.....	6
5. Startup.....	7
6. Operation Explanation for ELOG-MT (ADU mode)	9
6.1 File Selection Window.....	9
6.2 Time Series Window.....	10
6.3 Applied Filter Window	12
6.4 Power Spectra Window	13
6.5 Response Function Estimation Window.....	14
6.6 Apparent Resistivity and Phase Window.....	16
6.7 Vertical Magnetic Transfer Function (Tipper) Window.....	17
7. Operation Explanation for ELOG-MT (PHX mode).....	19
7.1 File Selection Window.....	19
7.2 Time Series Window.....	19
7.3 Applied Filter Window	21
7.4 Power Spectra Window	22
7.5 Response Function Estimation Window.....	23
7.6 Apparent Resistivity and Phase Window.....	26
7.7 Vertical Magnetic Transfer Function (Tipper) Window.....	27
8. Operation Explanation for ELOG-DUAL (ADU mode) + ELOG-MT	29
8.1 File Selection Window.....	29
8.2 Time Series Window.....	30
8.3 Applied Filter Window	32
8.4 Power Spectra Window	33
8.5 Response Function Estimation Window.....	34
8.6 Apparent Resistivity and Phase Window.....	36
9. Operation Explanation for ELOG-DUAL (ADU mode) + ATS	38
9.1 File Selection Window.....	38
9.2 Time Series Window.....	39

9.3	Applied Filter Window	41
9.4	Power Spectra Window	42
9.5	Response Function Estimation Window	43
9.6	Apparent Resistivity and Phase Window	46
10.	Operation Explanation for ELOG-DUAL (PHX mode)	48
10.1	File Selection Window	48
10.2	Time Series Window	49
10.3	Applied Filter Window	51
10.4	Power Spectra Window	52
10.5	Response Function Estimation Window	53
10.6	Apparent Resistivity and Phase Window	56
11.	Operation Explanation for ELOG1K + ELOG-MT	57
12.	Operation Explanation for ELOG1K + ATS	57
13.	Q & A	58
14.	Change History	62

1. Introduction

This software graphs ELOG-MT, ELOG1K, and ELOG-DUAL time-series data files on Windows. It supports low-pass, high-pass, and notch filters.

It also graphs the power spectrum of each channel.

This software enables one to estimate the impedance tensor and tipper from the time-series data. Robust estimators, prewhitening method, and the remote reference method are usable in the response function estimation.

Although this manual is for v1.1.0, some figures show screenshots from earlier versions when the displayed content is identical to the latest version.

2. License

This software is released under the BSD 3-Clause License.

Copyright (c) 2024, NT System Design

Redistribution and use in source and binary forms, with or without modification, are permitted provided that the following conditions are met:

1. Redistributions of source code must retain the above copyright notice, this list of conditions and the following disclaimer.
2. Redistributions in binary form must reproduce the above copyright notice, this list of conditions and the following disclaimer in the documentation and/or other materials provided with the distribution.
3. Neither the name of the copyright holder nor the names of its contributors may be used to endorse or promote products derived from this software without specific prior written permission.

THIS SOFTWARE IS PROVIDED BY THE COPYRIGHT HOLDERS AND CONTRIBUTORS "AS IS"

AND ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT

LIMITED TO, THE
IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A
PARTICULAR PURPOSE ARE
DISCLAIMED. IN NO EVENT SHALL THE COPYRIGHT HOLDER OR
CONTRIBUTORS BE LIABLE
FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR
CONSEQUENTIAL
DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF
SUBSTITUTE GOODS OR
SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS
INTERRUPTION) HOWEVER
CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT,
STRICT LIABILITY,
OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY
OUT OF THE USE
OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH
DAMAGE.

This software uses the following third parties. Among them, CLAPACK is used in
TRACMT.

1. TRACMT
2. ScottPlot
3. CLAPACK

The license terms of those third parties are described below.

TRACMT

<https://github.com/yoshiya-usui/TRACMT.git>

Copyright (c) 2024, Yoshiya Usui

Redistribution and use in source and binary forms, with or without
modification, are permitted provided that the following conditions are met:

1. Redistributions of source code must retain the above copyright notice, this

list of conditions and the following disclaimer.

2. Redistributions in binary form must reproduce the above copyright notice, this list of conditions and the following disclaimer in the documentation and/or other materials provided with the distribution.
3. Neither the name of the copyright holder nor the names of its contributors may be used to endorse or promote products derived from this software without specific prior written permission.

THIS SOFTWARE IS PROVIDED BY THE COPYRIGHT HOLDERS AND CONTRIBUTORS "AS IS"

AND ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE

IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE ARE

DISCLAIMED. IN NO EVENT SHALL THE COPYRIGHT HOLDER OR CONTRIBUTORS BE LIABLE

FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL

DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR

SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER

CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY,

OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE

OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

CLAPACK

<https://www.netlib.org/clapack/>

Copyright (c) 1992-2008 The University of Tennessee. All rights reserved.

\$COPYRIGHT\$

Additional copyrights may follow

\$HEADER\$

Redistribution and use in source and binary forms, with or without modification, are permitted provided that the following conditions are met:

- Redistributions of source code must retain the above copyright notice, this list of conditions and the following disclaimer.
- Redistributions in binary form must reproduce the above copyright notice, this list of conditions and the following disclaimer listed in this license in the documentation and/or other materials provided with the distribution.
- Neither the name of the copyright holders nor the names of its contributors may be used to endorse or promote products derived from this software without specific prior written permission.

THIS SOFTWARE IS PROVIDED BY THE COPYRIGHT HOLDERS AND CONTRIBUTORS

"AS IS" AND ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT

LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR

A PARTICULAR PURPOSE ARE DISCLAIMED. IN NO EVENT SHALL THE COPYRIGHT

OWNER OR CONTRIBUTORS BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL,

SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

ScottPlot

<https://github.com/ScottPlot/ScottPlot>

MIT License

Copyright (c) 2018 Scott Harden / Harden Technologies, LLC

Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files (the "Software"), to deal in the Software without restriction, including without limitation the rights to use, copy, modify, merge, publish, distribute, sublicense, and/or sell copies of the Software, and to permit persons to whom the Software is furnished to do so, subject to the following conditions:

The above copyright notice and this permission notice shall be included in all copies or substantial portions of the Software.

THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY,

FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE SOFTWARE.

3. Operating Environment

.NET Framework 6.0 running Windows 10 or higher.

CAUTION

This program may not work properly if the installation path or names of working folders contain non-ASCII characters (e.g., Japanese characters).

4. Installation

You should install .NET 6.0 Desktop Runtime and Visual C++ redistributable libraries. Please download from the following download sites and install them.

.NET 6.0 Desktop Runtime:

<https://dotnet.microsoft.com/en-us/download/dotnet/6.0>

Visual C++ redistributable libraries:

<https://learn.microsoft.com/en-us/cpp/windows/latest-supported-vc-redist?view=msvc-170>

You can install this software by copying all files under the “bin” folder to an install folder on your PC.

CAUTION

The name of the install folder and its parent folders should NOT contain non-ASCII characters (e.g., Japanese characters).

5. Startup

By just executing “ELOGFieldAnalyzer.exe”, you can launch the program. First, the “Select file type” window will be opened.

In this window, you can select the type of files to be read by this program. Explanations of the options are summarized in Table 1.

After selecting the file type, please press the “Select” button.

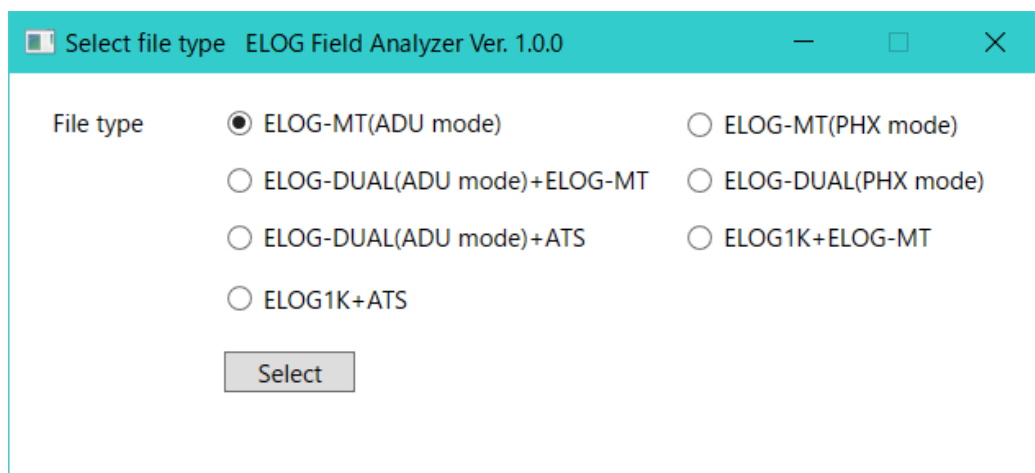


Figure 1: Select file type window

Table 1: Explanation of each file type

File type	Explanation
ELOG-MT (ADU mode)	Five-channel data of ELOG-MT (ADU mode) will be read.
ELOG-MT (PHX mode)	Five-channel data of ELOG-MT (PHX mode) will be read.
ELOG-DUAL (ADU mode) + ELOG-MT	Two-channel data of ELOG-DUAL (ADU mode) will be read. As an option, the magnetic field can also be read from ELOG-MT dat files and used to estimate response functions.
ELOG-DUAL (ADU mode) + ATS	Two-channel data of ELOG-DUAL (ADU mode) will be read. As an option, the magnetic field can also be read from ats files of Metronix instruments and used to estimate response functions.
ELOG-DUAL (PHX mode)	Two-channel data of ELOG-DUAL (PHX mode) will be read. As an option, the magnetic field can also be read from dat files of ELOG-MT and used to estimate response functions.
ELOG1K + ELOG-MT	Two-channel data of ELOG1K will be read. As an option, the magnetic field can also be read from dat files of ELOG-MT and used to estimate response functions.
ELOG1K + ATS	Two-channel data of ELOG1K will be read. As an option, the magnetic field can also be read from ats files of Metronix instruments and used to estimate response functions.

In what follows, the operation for each file type is explained.

6. Operation Explanation for ELOG-MT (ADU mode)

6.1 File Selection Window

Please select a folder where dat files of ELOG-MT are stored. This folder name must be a measurement day (YYYYMMDD).

You need to select the sampling frequency (32 or 1024 Hz) by the radio buttons on the right side.

If you check the box “Remote reference”, the magnetic field data is read from ats files of Metronix instruments as remote reference data (H_{rx} and H_{ry}). This reference data will be used in the remote reference method in the response estimation. You only need to select an ats file of the H_x component. The file of the H_y component is automatically selected in the program.

CAUTION

The ats file of the H_x component must contain “_C02_” as the channel index and “_THx_.” Otherwise, the program cannot read the ats file.

The times during which the reference magnetic field was measured should overlap with the measurement time of the ELOG-MT at least partly.

Please press the “Read files” button after you properly select sampling frequency and folder and file names.

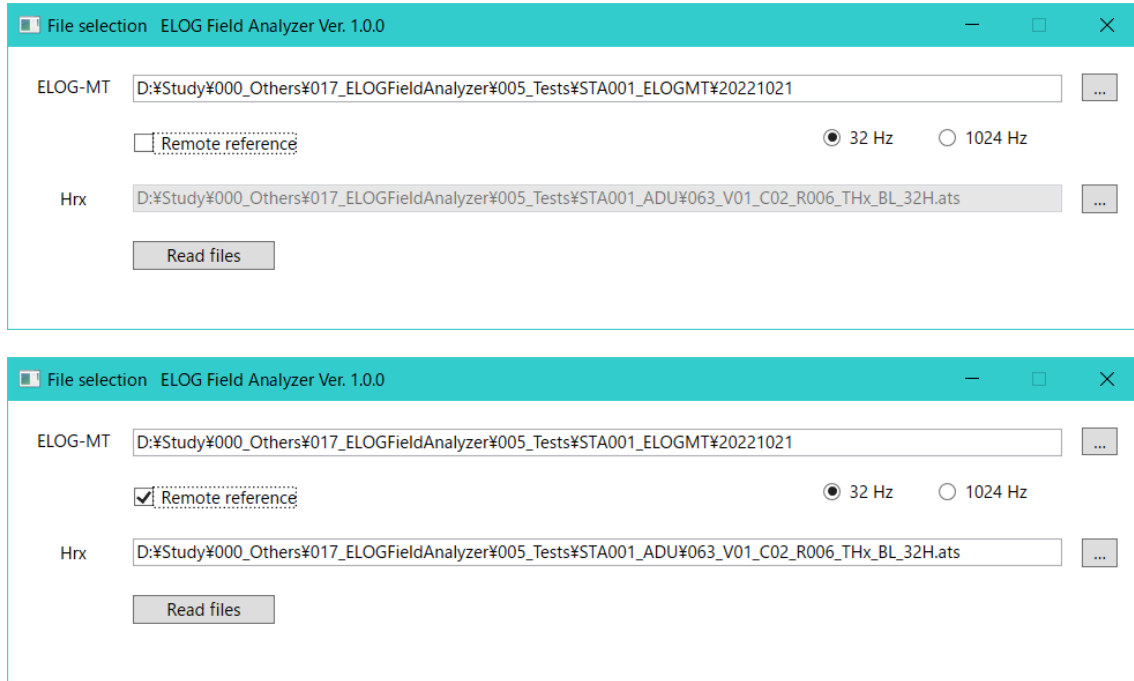


Figure 2: File selection window

6.2 Time Series Window

If dat files (and ats files) are appropriately read, time-series data are displayed.

The explanation of the items in this window (marked in Figure 3) is written below.

EXPLANATION OF EACH ITEM

- (1) The time that the cursor points to is shown in the title bar.
- (2) The time series of each channel is displayed. If you double-click the window, the time series will be displayed in a new window, the size of which is changeable.
- (3) Star time of the time series displayed in the window.
- (4) End time of the time series displayed in the window.
- (5) Button to show the tail end of the time series of the previous day.
- (6) Button to show the time series of the previous day in the same time range.
- (7) Button to show the time series of the next day in the same time range.
- (8) Button to show the head of the time series of the next day.
- (9) Button to decrement the time range. It is not active if the “Type” is “All”.
- (10) Slide bar to change the time range. It is not active if the “Type” is “All”.
- (11) Button to increment the time range. It is not active if the “Type” is “All”.
- (12) Button to apply a low-pass filter.

- (13) Cutoff frequency (Hz) of the low-pass filter.
- (14) Button to apply a high-pass filter.
- (15) Cutoff frequency (Hz) of the high-pass filter.
- (16) Button to apply a notch filter.
- (17) Cutoff frequencies (Hz) of the notch filter. You can write multiple frequencies, comma delimited.
- (18) Clear all digital filters.
- (19) Button to show the information of the applied digital filters.
- (20) Type of how to select the time range. If you select “All”, all of the time series of the specified day are displayed. If “Data number” is selected, the number of data is controlled by the selection in (21). If “Data span” is selected, the time span is controlled by the selection in (22).
- (21) The number of data shown in this window. This control is active only if “Type” is “Data number”. If you change the number, “Type” is automatically changed to “Data number.”
- (22) The time span of the time series shown in this window. This control is active only if “Type” is “Data span”. If you change the time span, “Type” is automatically changed to “Data span.”
- (23) Button to show the power spectra.
- (24) Button to go to the windows for the response function estimation.

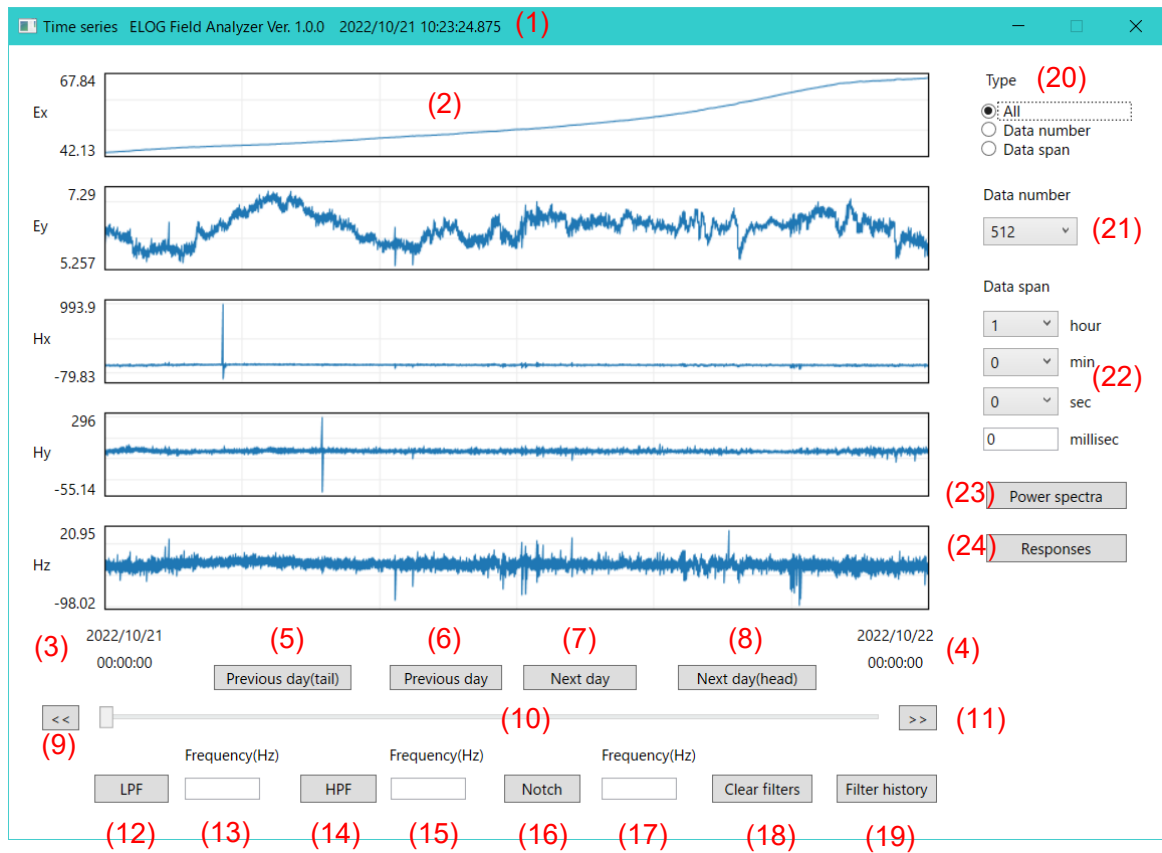


Figure 3: Time series window

6.3 Applied Filter Window

When you press the “Filter history” button, the “Applied filter” window is opened. In this window, the amplitude characteristics of the digital filters you already applied are shown (characteristics of all the filters you applied are included).

The information on the filters you have applied is listed in the lower left panel.

You can change the range of the upper figure by pressing the “Change range” button after specifying the upper and lower bounds. If you press this button with a blank textbox. Upper and/or lower bounds are automatically determined.

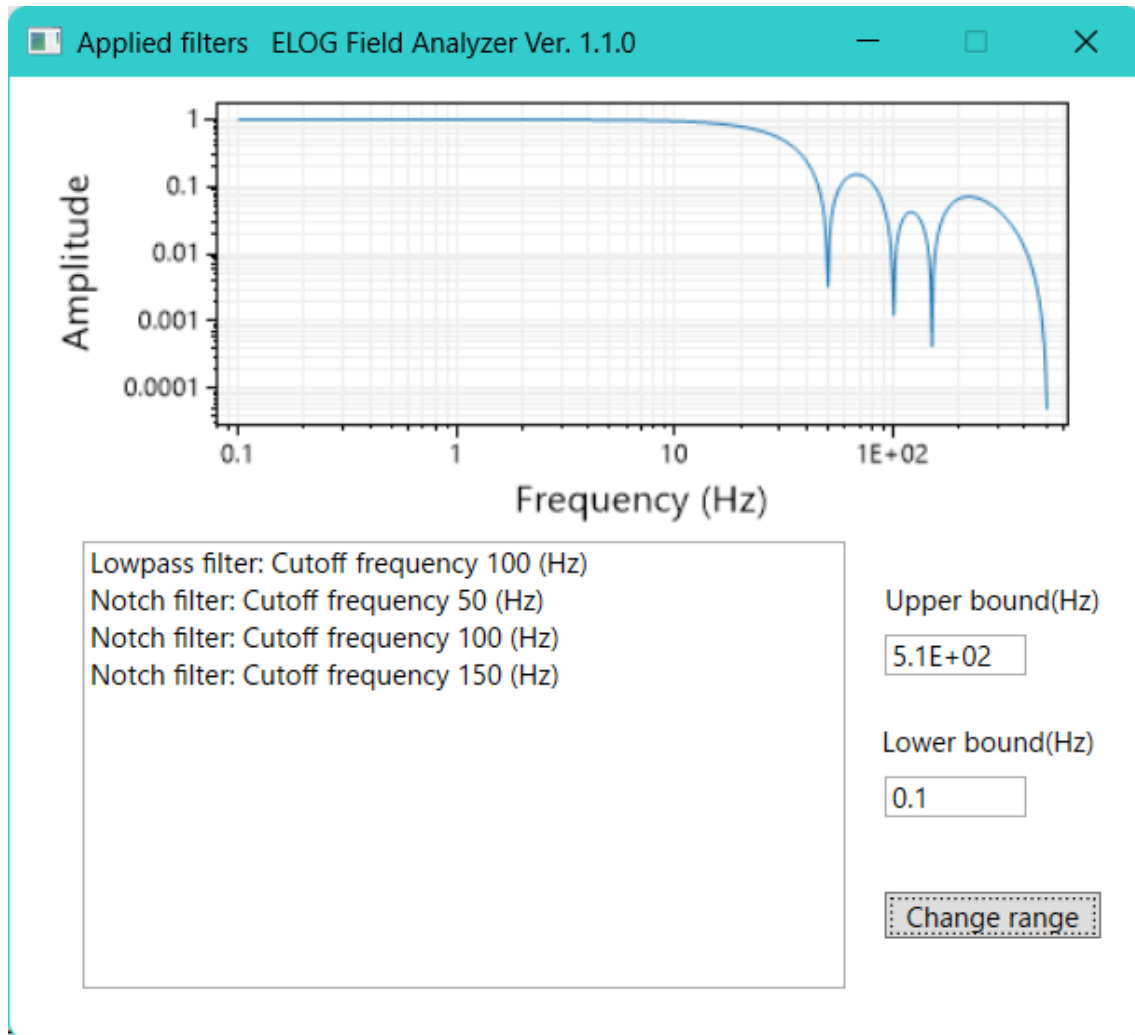


Figure 4: Applied filters window

6.4 Power Spectra Window

When you press the “Power spectra” button, the “Power spectra” window opens. In this window, the power spectra of all channels are displayed.

Only the data of the times displayed in the “Time series” window are used for calculating the power spectra.

The explanation of the items in this window (marked in Figure 5) is written below.

EXPLANATION OF EACH ITEM

- (1) The power spectrum of each channel is displayed. If you double-click the window, the power spectrum opens in a new window, whose size is adjustable.
- (2) The minimum frequency displayed.
- (3) The maximum frequency displayed.

- (4) If you check the box, the linear trend in the logarithmic axis is removed from each power spectrum. If you uncheck, the original power spectra are redisplayed.
- (5) Lower limit of the frequencies to be displayed.
- (6) Upper limit of the frequencies to be displayed.
- (7) The button to change the range of the horizontal axes. If you press the button, the range of the frequency is bounded based on the selection in (5) and (6). If (5) and/or (6) are blank, the range is automatically determined.
- (8) The frequency that the cursor points to is shown here.

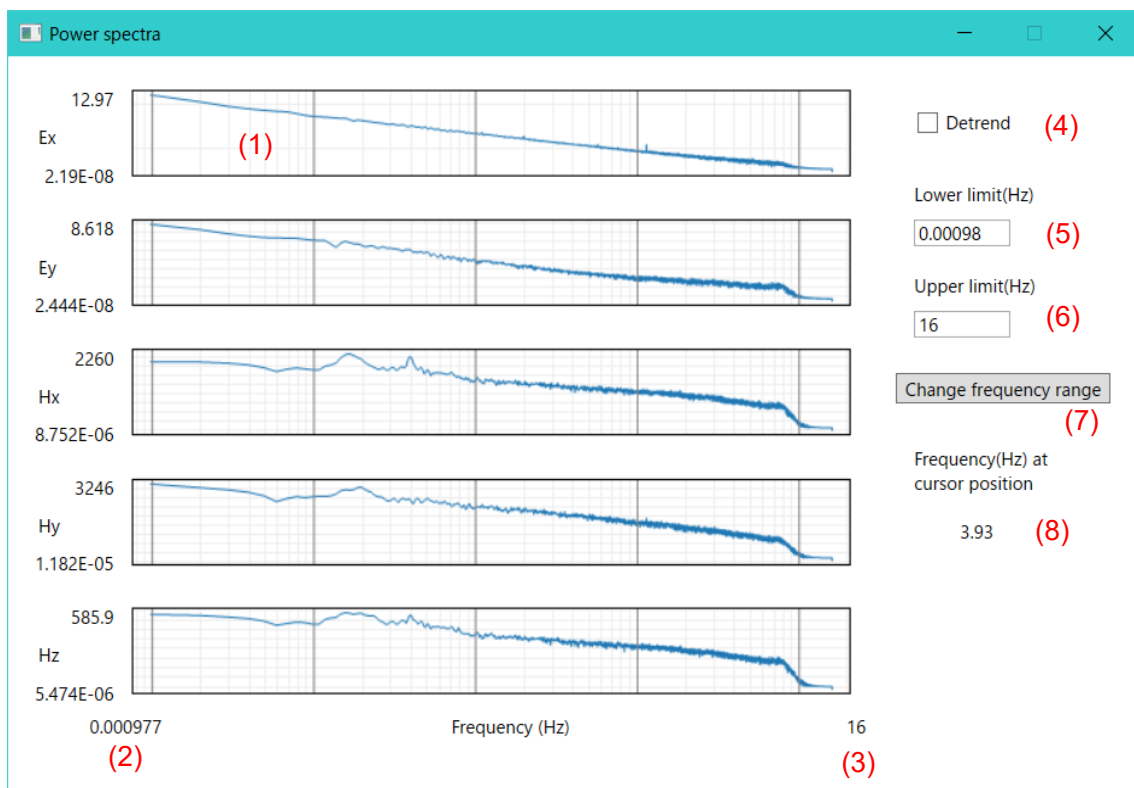


Figure 5: Power spectra window

6.5 Response Function Estimation Window

When you press the "Reponses" button, the "Response function estimation" window is opened. In this window, you can select the options for the estimation of the response functions, start the estimation, and plot the sounding curves of the estimated response functions.

Only the data of the times displayed in the "Time series" window are used for the response function estimation.

The explanation of the items on this window (marked in Figure 6) is written below.

EXPLANATION OF EACH ITEM

- (1) Work folder. You need to select the work folder for the response function estimation.
- (2) Type of the response functions to be estimated.
- (3) Dipole length for the N-S component of the electric field (unit: meter).
- (4) Dipole length for the E-W component of the electric field (unit: meter).
- (5) Measurement direction from the north (unit: degree). For example, if the +x direction corresponds to the geomagnetic north in the measurement, and the declination around the station is 7 degrees to the west, one needs to write -7 (degrees) in this textbox.
- (6) Rotation angle to the coordinate system in which the response functions are estimated (unit: degree). For example, if you write 40 (degrees) in the textbox, the response functions are estimated in the coordinate system whose +x direction is N40°E.
- (7) Coil calibration file for the H_x component. This file should follow Metronix's calibration file format.
- (8) Coil calibration file for the H_y component. This file should follow Metronix's calibration file format.
- (9) Coil calibration file for the H_z component. This file should follow Metronix's calibration file format.
- (10) Estimation method. OLS means the ordinary least square method. OLS can compute response functions most quickly. If you select “M-estimator” or “RRMS”, outliers are down-weighted. The RRMS estimator is generally the most robust but also the most time-consuming. “RRMS” can be selected only when the remote magnetic field data are read.
- (11) If you check it, the ordinary prewhitening method is used, and the response functions are estimated at longer periods than when it is unchecked. This is because the Fourier transforms at the third and fourth frequencies are used if the prewhitening is applied (because the prewhitening can suppress the spectral leakage), whereas the Fourier transforms at the eighth and twelfth fourth frequencies are used when no prewhitening is applied.
- (12) If you check it, the original time series data are downsampled.
- (13) Rate of downsampling. For example, if you wrote 32 for 32 Hz sampling data, the original time series data are downsampled to 1 Hz sampling data.
- (14) Button to make an input file of TRACMT. The input file is created in the work folder specified in (1). When the input file is successfully made, a message appears at the bottom of the window.

- (15) Button to open the created input file with a text editor. You can check and modify the input file with the text editor.
- (16) Button to start TRACMT to estimate the response function. If you press the button, a command prompt launches, in which the message from TRACMT is displayed.
- (17) Button to plot the sounding curves of the estimated response functions.

Response function estimation ELOG Field Analyzer Ver. 1.0.0

Work folder (1) D:\temp\work ...

Response type (2) ☒ Impedance tensor ☐ Tipper ☐ Impedance tensor + Tipper

Dipole length(m) N-S (3) 30 E-W (4) 30 Measurement direction from the north (deg) (5) 0 Rotation (deg) (6) 0

Coil calibration files

Hx (7) D:\Study\000_Others\017_ELOGFieldAnalyzer\005_Tests\STA002_ADU\MFS06390_20171020R.TXT ...

Hy (8) D:\Study\000_Others\017_ELOGFieldAnalyzer\005_Tests\STA002_ADU\MFS06391_20171020R.TXT ...

Hz (9) D:\Study\000_Others\017_ELOGFieldAnalyzer\005_Tests\STA002_ADU\MFS07e676_20220627.TXT ...

Estimation method (10) ☒ OLS ☐ M-estimator ☐ RRMS ☐ Prewhitening (11) ☐ Downsampling (12) 1/ (13)

(14) Make input file (15) Edit input file (16) Start calculation (17) Plot responses

Figure 6: Response function estimation window

6.6 Apparent Resistivity and Phase Window

If you press the “Plot responses” button in the “Response function estimation” window after the calculation of TRACMT is finished, the “Apparent resistivity and phase” window is opened.

In this window, the squared coherence and the sounding curves of the apparent resistivity and phase are depicted.

CAUTION

You need to select “Impedance tensor” or “Impedance tensor + Tipper” in “Response type” to calculate the impedance tensor.

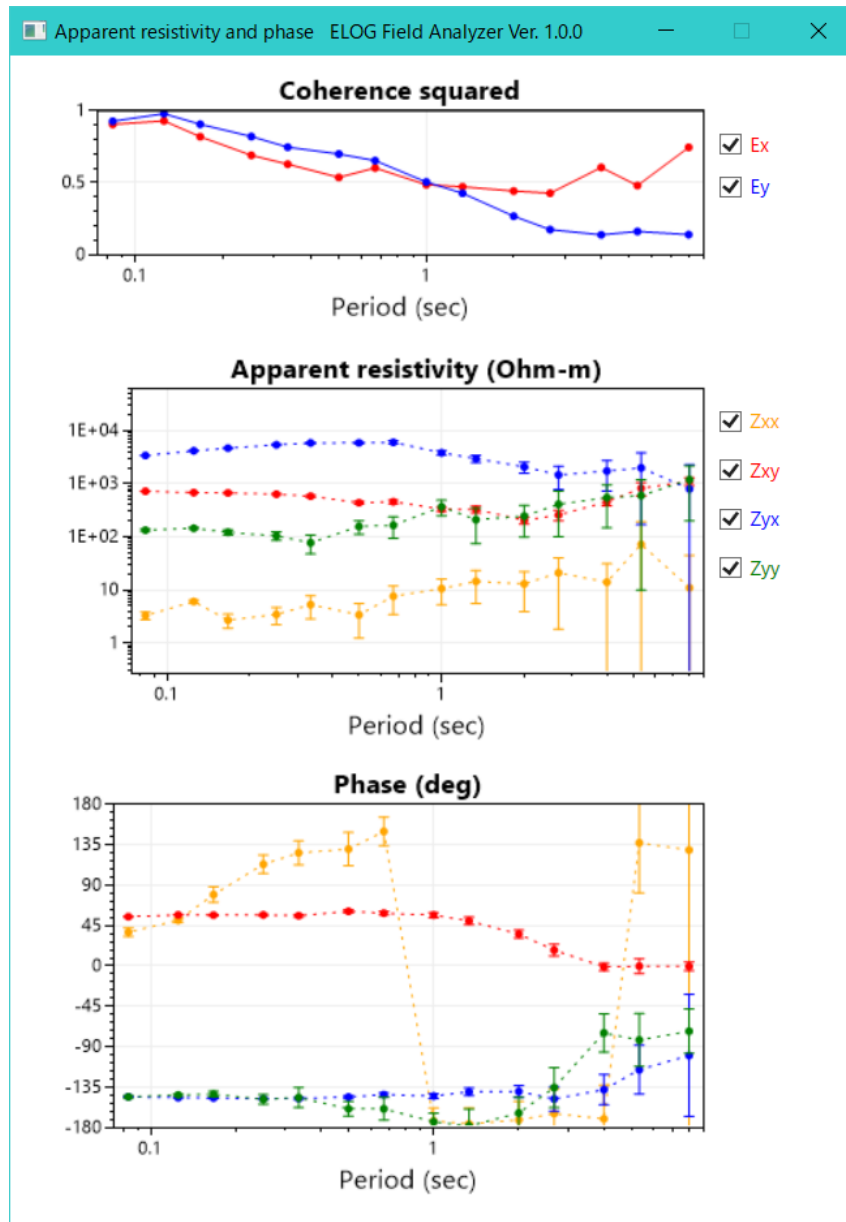


Figure 7: Apparent resistivity and phase window

6.7 Vertical Magnetic Transfer Function (Tipper) Window

If you press the “Plot responses” button in the “Response function estimation” window after the calculation of TRACMT is finished, the “Vertical magnetic transfer function (tipper)” window is opened.

In this window, the squared coherence and the vertical magnetic transfer function (tipper) are depicted.

CAUTION

You need select “Tipper” or “Impedance tensor + Tipper” in “Response type” to calculate the impedance tensor.

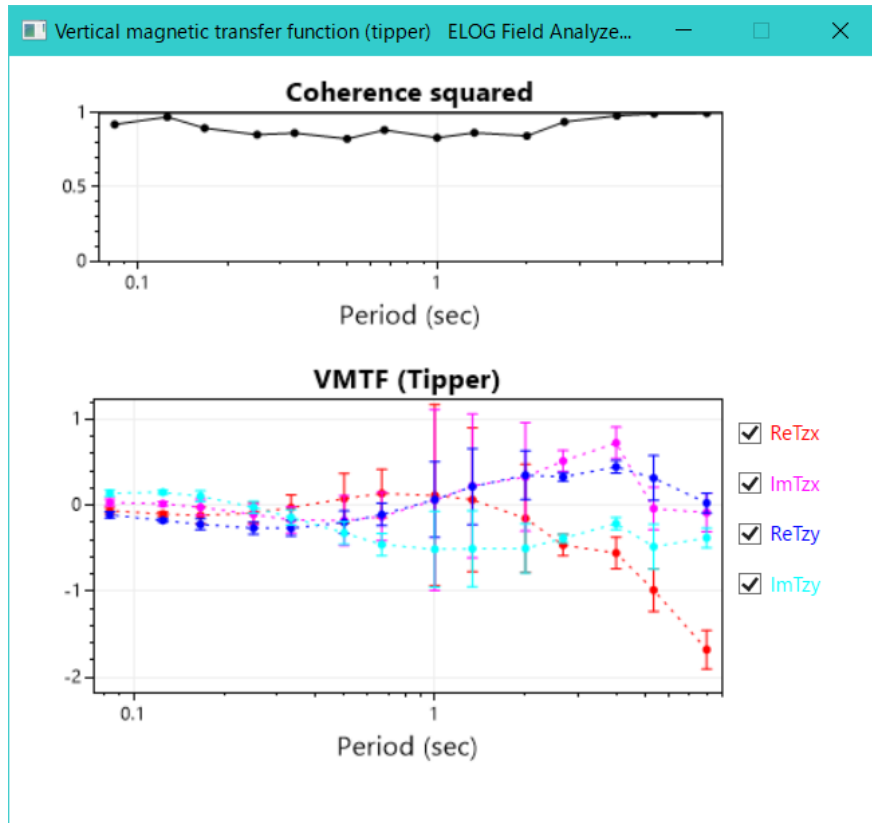


Figure 8: Vertical magnetic transfer function (tipper) window

7. Operation Explanation for ELOG-MT (PHX mode)

7.1 File Selection Window

Please select a folder where dat files of ELOG-MT are stored. This folder name must be the measurement day (YYYYMMDD).

You need to select the sampling frequency (15, 150, or 2400 Hz) by the radio buttons on the right side.

If you check the box “Remote reference”, the magnetic field data is read from dat files of another ELOG-MT station as remote reference data (H_{rx} and H_{ry}). This reference data will be used in the remote reference method in the response estimation.

CAUTION

The time period of the local and remote stations should be overlapped at least partly.

Please press the “Read files” button after you properly select sampling frequency and folder and file names.

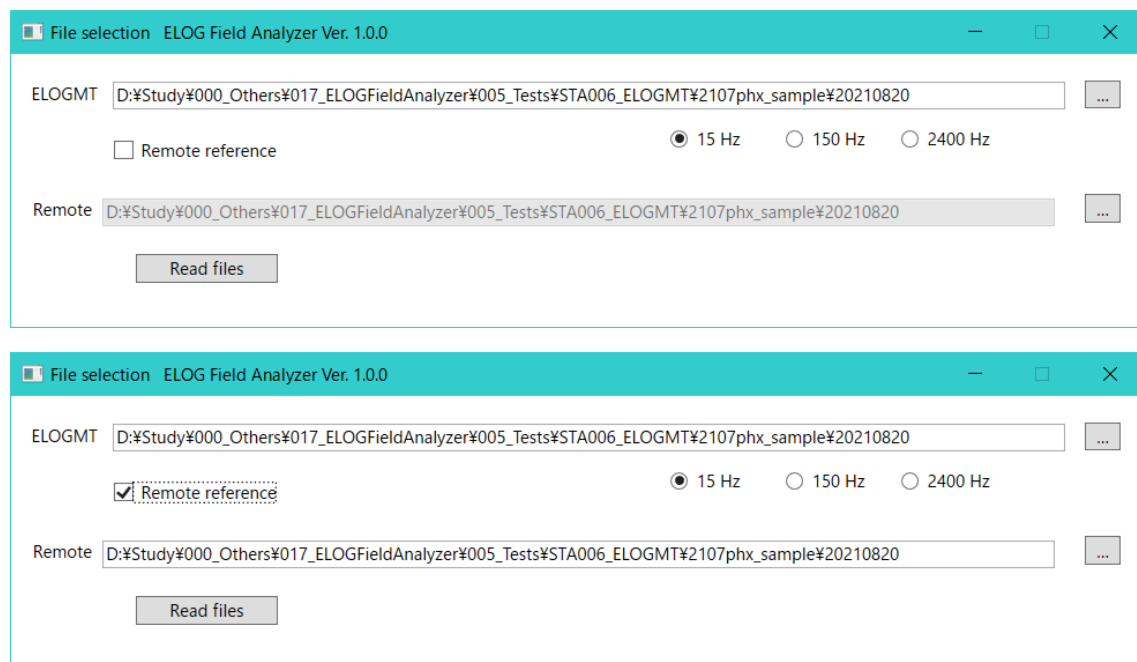


Figure 9: File selection window

7.2 Time Series Window

If dat files are appropriately read, time-series data are displayed.

The explanation of the items on this window (marked in Figure 10) is written below.

EXPLANATION OF EACH ITEM

- (1) The time that the cursor points to is shown in the title bar.
- (2) The time series of each channel is displayed. If you double-click the window, the time series will be displayed in a new window, the size of which is changeable.
- (3) Star time of the time series displayed in the window.
- (4) End time of the time series displayed in the window.
- (5) Button to show the tail end of the time series of the previous day.
- (6) Button to show the time series of the previous day in the same time range.
- (7) Button to show the time series of the next day in the same time range.
- (8) Button to show the head of the time series of the next day.
- (9) Button to decrement the time range. It is not active if the “Type” is “All”.
- (10) Slide bar to change the time range. It is not active if the “Type” is “All”.
- (11) Button to increment the time range. It is not active if the “Type” is “All”.
- (12) Button to apply a low-pass filter.
- (13) Cutoff frequency (Hz) of the low-pass filter.
- (14) Button to apply a high-pass filter.
- (15) Cutoff frequency (Hz) of the high-pass filter.
- (16) Button to apply a notch filter.
- (17) Cutoff frequencies (Hz) of the notch filter. You can write multiple frequencies, comma delimited.
- (18) Clear all digital filters applied.
- (19) Button to show the information on the applied digital filters.
- (20) If you select “All”, all of the time series of the specified day are displayed. If “Data number” is selected, the number of data is controlled by the selection in (21). If “Data span” is selected, the time span is controlled by the selection in (22).
- (21) The number of data shown in this window. This control is active only if “Type” is “Data number”. If you change the number, “Type” is automatically changed to “Data number.”
- (22) The time span of the time series shown in this window. This control is active only if “Type” is “Data span”. If you change the time span, “Type” is automatically changed to “Data span.”
- (23) Button to show the power spectra.

(24) Button to go to the windows for the response function estimation.

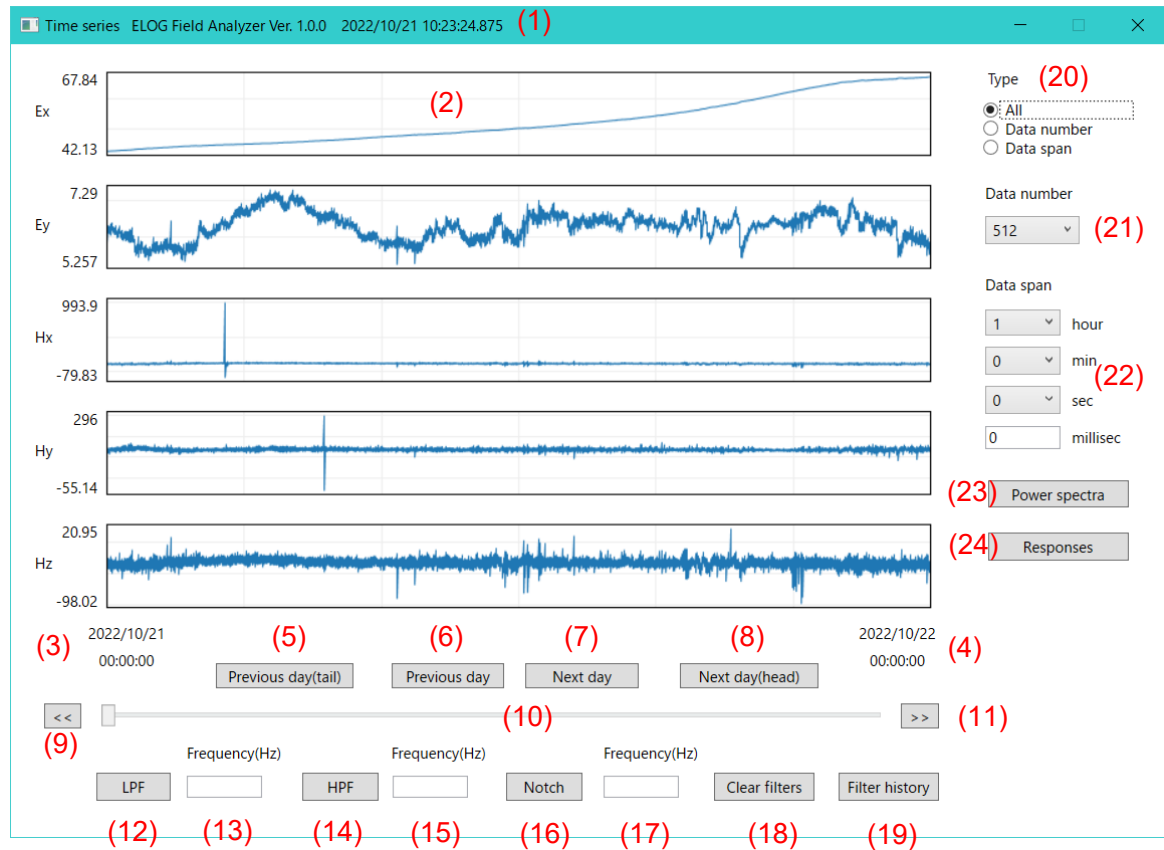


Figure 10: Time series window

7.3 Applied Filter Window

When you press the “Filter history” button, the “Applied filter” window will be opened. In this window, the amplitude characteristics of the digital filters you applied are shown (characteristics of all the filters you applied are merged).

The information on the filters you have applied is listed in the lower left panel.

You can change the range of the upper figure by pressing the “Change range” button after specifying the upper and lower bounds. If you press this button with a blank textbox. Upper and/or lower bounds are automatically determined.

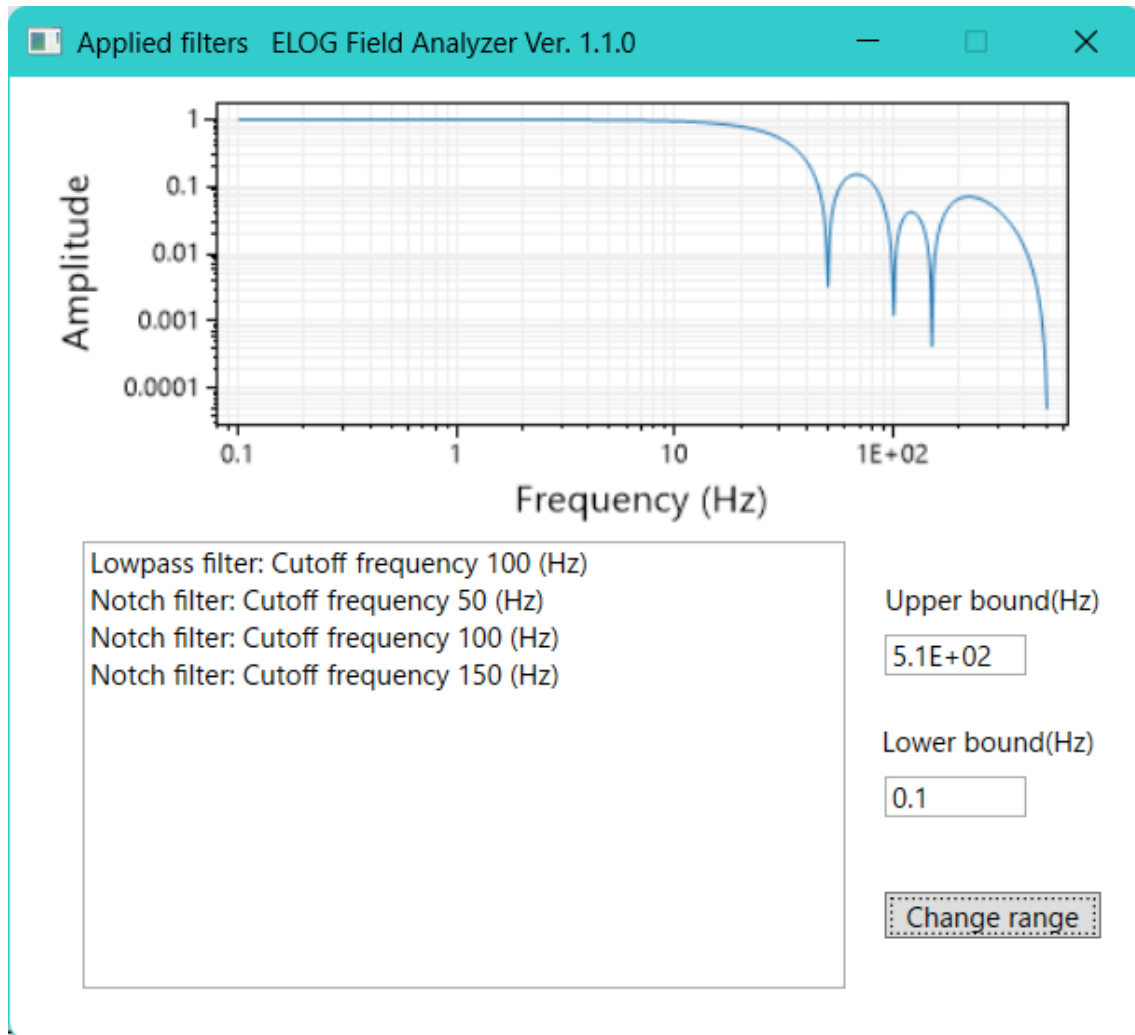


Figure 11: Applied filters window

7.4 Power Spectra Window

When you press the “Power spectra” button, the “Power spectra” window opens. In this window, the power spectra of all channels are displayed.

Only the data of the times displayed in the “Time series” window are used for calculating the power spectra.

The explanation of the items in this window (marked in Figure 12) is written below.

EXPLANATION OF EACH ITEM

- (1) The power spectrum of each channel is displayed. If you double-click the window, the power spectrum will be displayed in a new window. whose size is adjustable.
- (2) The minimum frequency displayed.
- (3) The maximum frequency displayed.

- (4) If you check the box, the linear trend in the logarithmic axis is removed from each power spectrum. If you uncheck, the original power spectra are redisplayed.
- (5) Lower limit of the frequencies to be displayed.
- (6) Upper limit of the frequencies to be displayed.
- (7) The button to change the range of the horizontal axes. If you press the button, the range of the frequency is bounded based on the selection in (5) and (6). If (5) and/or (6) are blank, the range is automatically determined.
- (8) The frequency that the cursor points to is shown here.

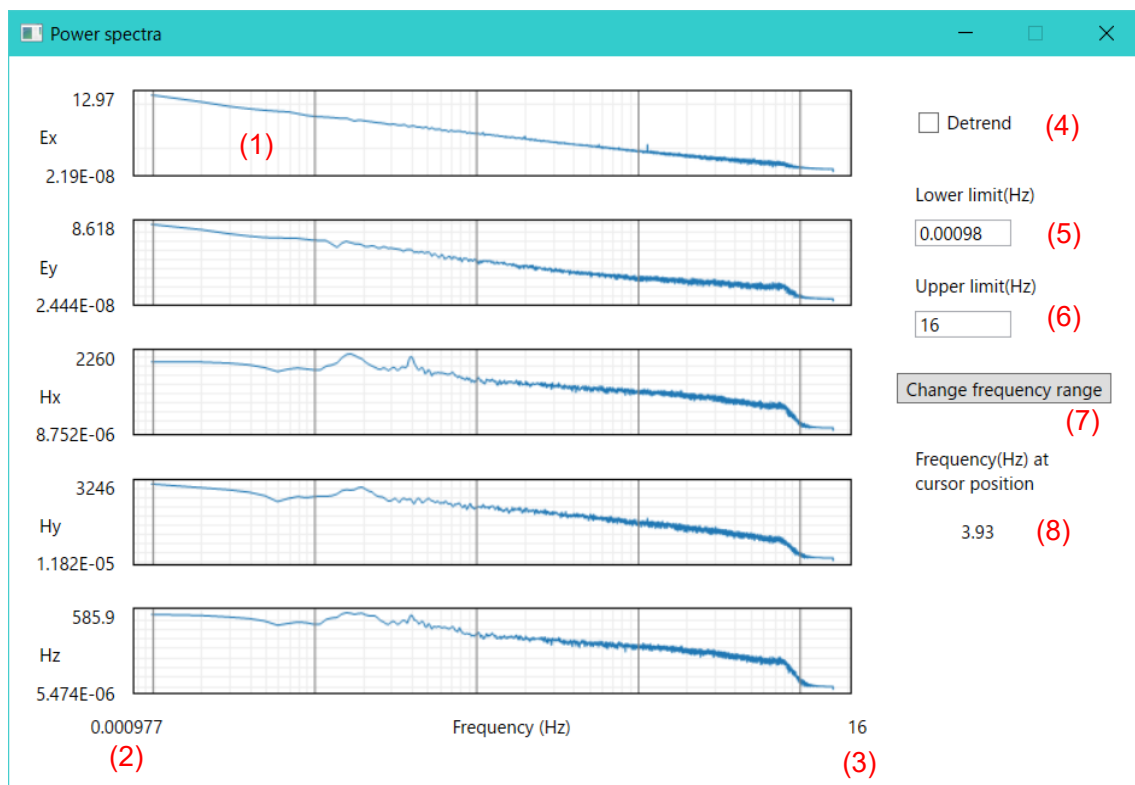


Figure 12: Power spectra window

7.5 Response Function Estimation Window

When you press the "Reponses" button, the "Response function estimation" window is opened. In this window, you can select options in the estimation of the response functions, start the estimation, and plot the sounding curves of the estimated response functions.

Only the data of the times displayed in the "Time series" window are used for the response function estimation

The explanation of the items in this window (marked in Figure 13) is written below.

EXPLANATION OF EACH ITEM

- (1) Work folder. You need to select the work folder for the response function estimation.
- (2) Type of the response functions to be estimated.
- (3) Dipole length for the N-S component of the electric field (unit: meter).
- (4) Dipole length for the E-W component of the electric field (unit: meter).
- (5) Measurement direction from the north (unit: degree). For example, if the +x direction corresponds to the geomagnetic north in the measurement, and the declination around the station is 7 degrees to the west, one needs to write -7 (degrees) in this textbox.
- (6) Rotation angle to the coordinate system in which the response functions are estimated (unit: degree). For example, if you write 40 (degrees) in the textbox, the response functions are estimated in the coordinate system whose +x direction is N40°E.
- (7) Coil calibration file for the Hx component. The file should follow the format described below.
- (8) Coil calibration file for the Hy component. The file should follow the format described below.
- (9) Coil calibration file for the Hz component. The file should follow the format described below.
- (10) Estimation method. OLS means the ordinary least square method. OLS can compute response functions most quickly. If you select “M-estimator” or “RRMS”, outliers are down-weighted. The RRMS estimator is generally the most robust but also the most time-consuming. “RRMS” can be selected only when the remote magnetic field data are read.
- (11) If you check this, the ordinary prewhitening method is used, and the response functions are estimated at longer periods than when it is unchecked. This is because the Fourier transforms at the third and fourth frequencies are used if the prewhitening is applied (because the prewhitening can suppress the spectral leakage), whereas the Fourier transforms at the eighth and twelfth frequencies are used when no prewhitening is applied.
- (12) If you check this, the original time series data are downsampled.
- (13) Rate of downsampling. For example, if you wrote 15 for 15 Hz sampling data, the original time series data are downsampled to 1 Hz sampling data.
- (14) Button to make an input file of TRACMT. The input file is created in the work folder you specify in (1). When the input file is successfully made, a message appears at the bottom of the window.

- (15) Button to open the created input file with a text editor. You can check and modify the input file by the text editor.
- (16) Button to start TRACMT to estimate the response function. If you press the button, a command prompt launches, in which the message from TRACMT is displayed.
- (17) Button to plot the sounding curves of estimated response functions.

The screenshot shows the 'Response function estimation' window. It includes a 'Work folder' field (1), 'Response type' radio buttons (2) for 'Impedance tensor', 'Tipper', and 'Impedance tensor + Tipper'. Below are 'Dipole length (m)' fields for 'N-S' (3) and 'E-W' (4), and 'Measurement direction from the north (deg)' (5) and 'Rotation (deg)' (6) fields. The 'Coil calibration files' section has fields for 'Hx' (7), 'Hy' (8), and 'Hz' (9). The 'Estimation method' section has radio buttons for 'OLS' (10), 'M-estimator', and 'RRMS', and checkboxes for 'Prewhitening' (11) and 'Downsampling' (12). At the bottom are four buttons: 'Make input file' (14), 'Edit input file' (15), 'Start calculation' (16), and 'Plot responses' (17). A '1/' field (13) is also present next to the 'Downsampling' checkbox.

Figure 13: Response function estimation window

FORMAT OF COIL CALIBRATION FILES

Coil calibration files for the magnetic field should follow the following format.

Instrumental characteristics are corrected in the frequency domain by multiplying A and $C(f)$ as

$$A \times C(f_1)H_x(f_1) \rightarrow H_x(f_1)$$

$$A \times C(f_2)H_x(f_2) \rightarrow H_x(f_2)$$

...

The calibration function is interpolated for the frequency at which $C(f)$ is not specified.

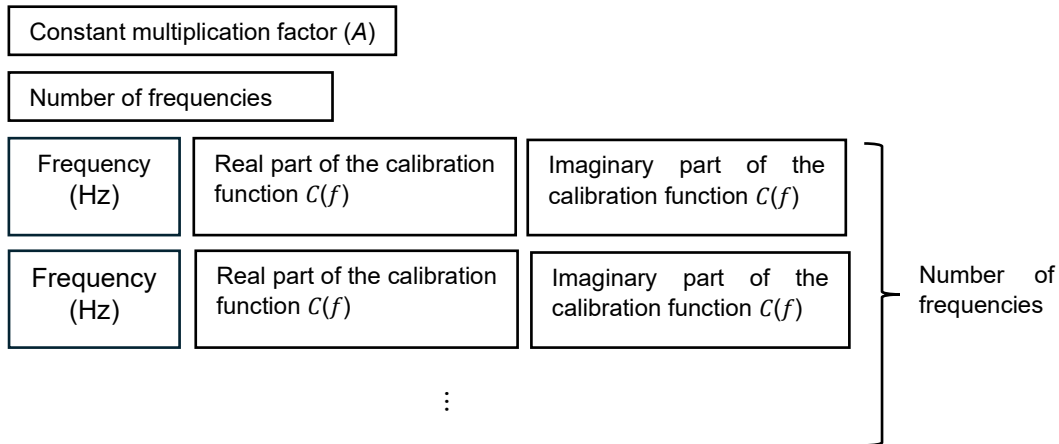


Figure 14: Format of calibration file of TRACMT

7.6 Apparent Resistivity and Phase Window

If you press the “Plot responses” button in the “Response function estimation” window after the calculation of TRACMT is finished, the “Apparent resistivity and phase” window is opened.

In this window, the squared coherence and the sounding curves of the apparent resistivity and phase are depicted.

CAUTION

You need to select “Impedance tensor” or “Impedance tensor + Tipper” in “Response type” to calculate the impedance tensor.

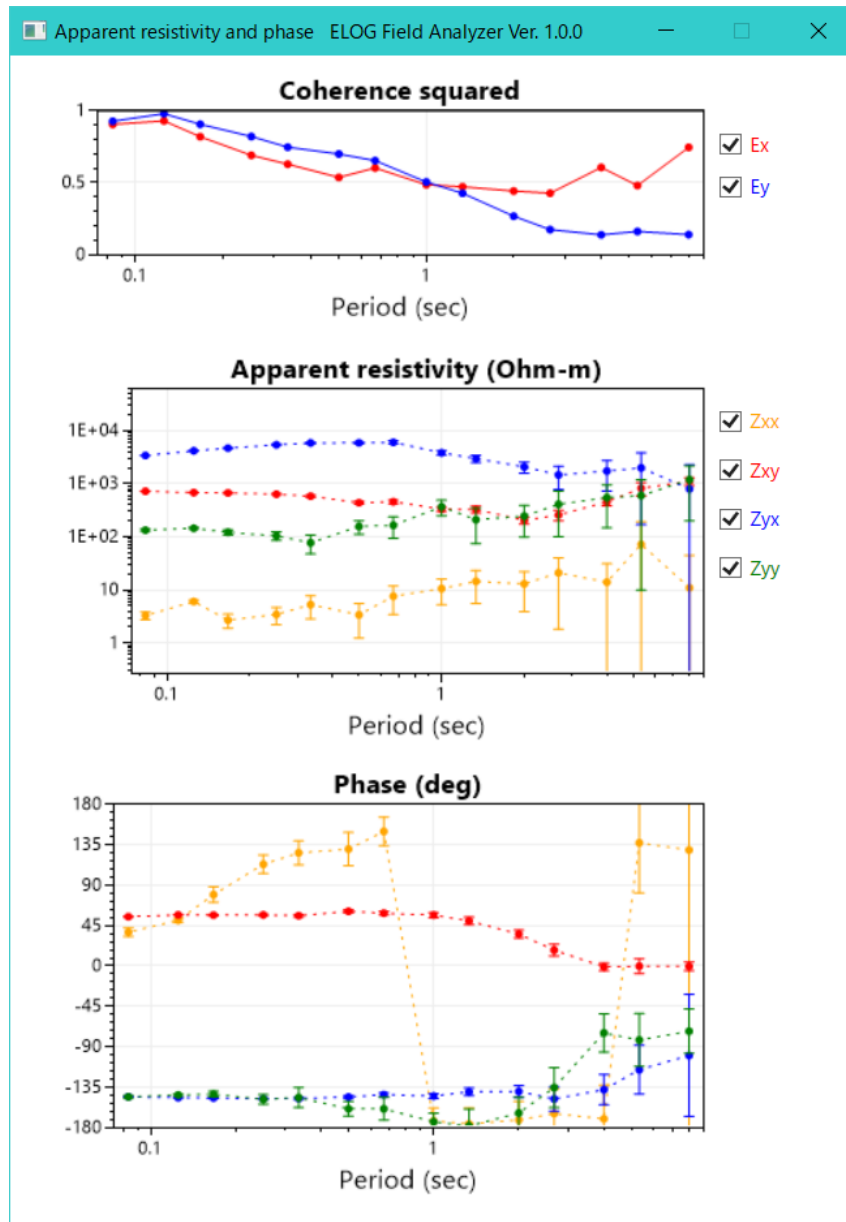


Figure 15: Apparent resistivity and phase window

7.7 Vertical Magnetic Transfer Function (Tipper) Window

If you press the “Plot responses” button in the “Response function estimation” window after the calculation of TRACMT is finished, the “Vertical magnetic transfer function (tipper)” window is opened.

In this window, the squared coherence and the vertical magnetic transfer function (tipper) are depicted.

CAUTION

You need to select “Tipper” or “Impedance tensor + Tipper” in “Response type” to calculate the impedance tensor.

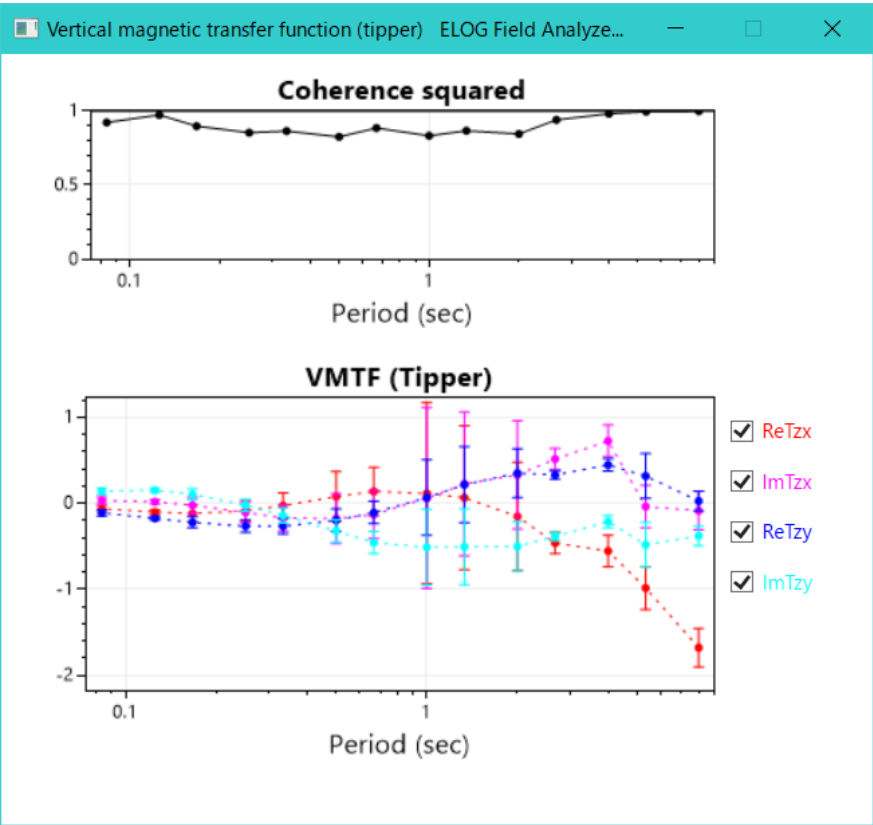


Figure 16: Vertical magnetic transfer function (tipper) window

8. Operation Explanation for ELOG-DUAL (ADU mode) + ELOG-MT

8.1 File Selection Window

Please select a folder where dat files of ELOG-DUAL are stored. This folder name must be the measurement day (YYYYMMDD).

You need to select the sampling frequency (32 or 1024 Hz) by the radio buttons on the right side.

If you check the box “Magnetic field data,” the horizontal components (H_x and H_y) of the magnetic field are read from dat files of an ELOG-MT station. These magnetic field data are used to estimate response functions.

If you check the box “Remote reference”, the magnetic field data is read from ats files of Metronix instruments as remote reference data (H_{rx} and H_{ry}). This reference data will be used in the remote reference method in the response estimation. You only need to select the ats file for the H_x component. The file for the H_y component is automatically selected in the program.

CAUTION

The measurement times of all channels should overlap at least partly.

The ats file of the H_x component must contain “_C02_” as the channel index and “_THx_.” Otherwise, the program cannot read the ats file.

Please press the “Read files” button after you properly select the sampling frequency and the folder and file names.

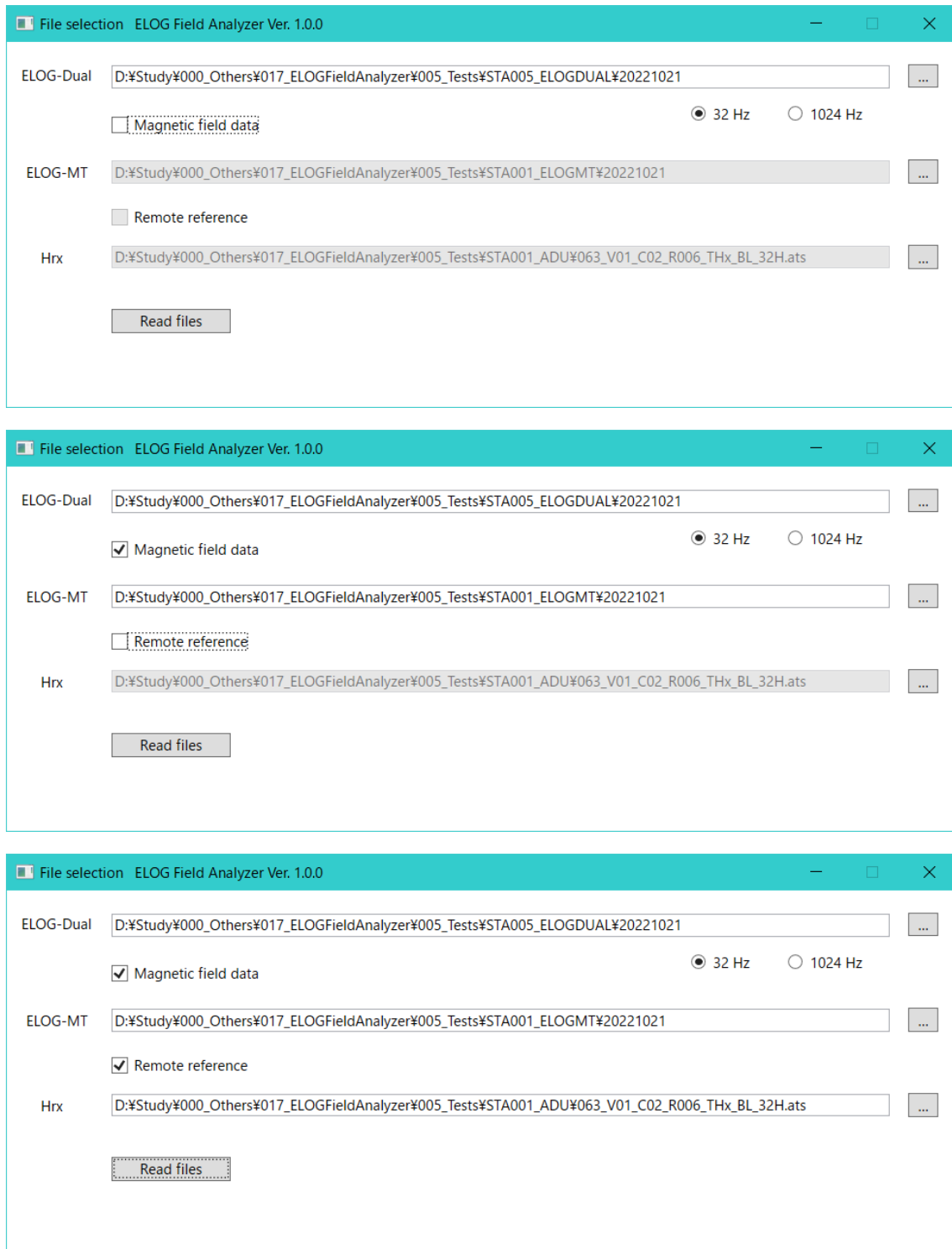


Figure 17: File selection window

8.2 Time Series Window

If dat files are appropriately read, time-series data are displayed.

The explanation of the items in this window (marked in Figure 18) is written below.

EXPLANATION OF EACH ITEM

- (1) The time that the cursor points to the window is shown in the title bar.
- (2) The time series of each channel is displayed. If you double-click the window, the time series will be displayed in a new window, the size of which is changeable.
- (3) Star time of the time series displayed in the window.
- (4) End time of the time series displayed in the window.
- (5) Button to show the tail end of the time series of the previous day.
- (6) Button to show the time series of the previous day in the same time range.
- (7) Button to show the time series of the next day in the same time range.
- (8) Button to show the head of the time series of the next day.
- (9) Button to decrement the time range. It is not active if the “Type” is “All”.
- (10) Slide bar to change the time range. It is not active if the “Type” is “All”.
- (11) Button to increment the time range. It is not active if the “Type” is “All”.
- (12) Button to apply a low-pass filter.
- (13) Cutoff frequency (Hz) of the low-pass filter.
- (14) Button to apply a high-pass filter.
- (15) Cutoff frequency (Hz) of the high-pass filter.
- (16) Button to apply a notch filter.
- (17) Cutoff frequencies (Hz) of the notch filter. You can write multiple frequencies, comma delimited.
- (18) Clear all digital filters applied.
- (19) Button to show the information on the applied digital filters.
- (20) If you select “All”, all of the time series of the specified day are displayed. If “Data number” is selected, the number of data is controlled by the selection in (21). If “Data span” is selected, the time span is controlled by the selection in (22).
- (21) The number of data shown in this window. This control is active only if “Type” is “Data number”. If you change the number, “Type” is automatically changed to “Data number.”
- (22) The time span of the time series shown in this window. This control is active only if “Type” is “Data span”. If you change the time span, “Type” is automatically changed to “Data span.”
- (23) Button to show the power spectra.

(24) Button to go to the windows for the response function estimation.

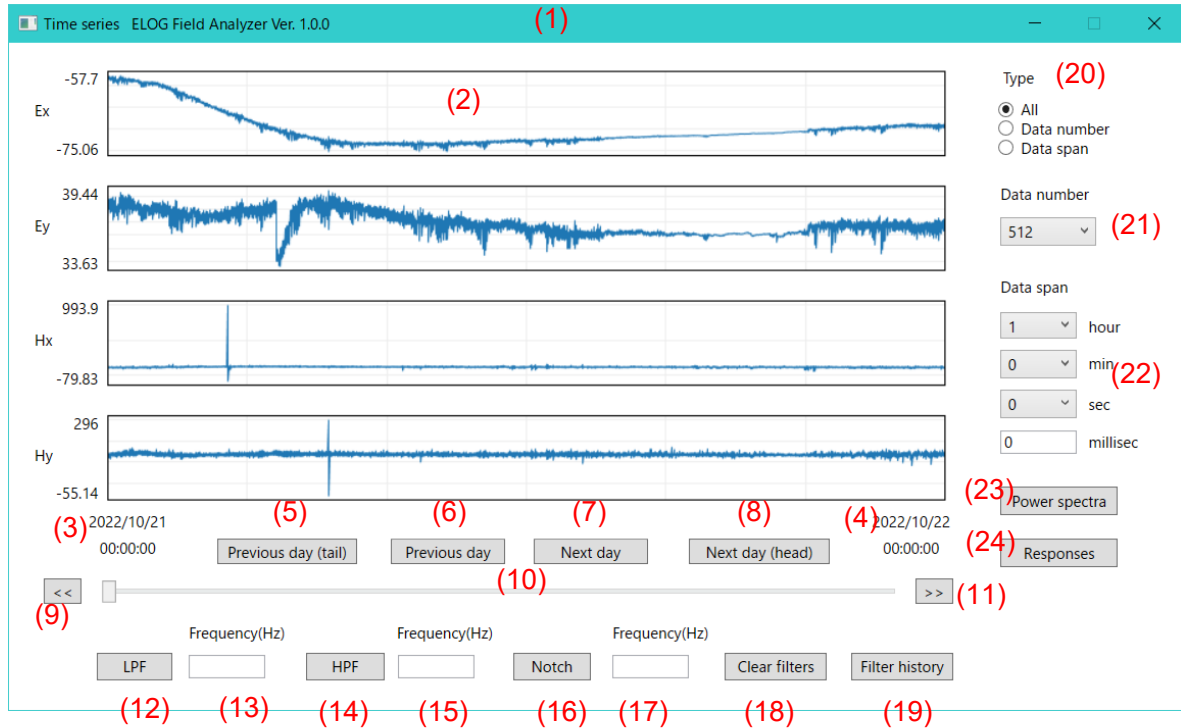


Figure 18: Time series window (in the case where the reference magnetic field is not read)

8.3 Applied Filter Window

When you press the “Filter history” button, the “Applied filter” window will be opened. In this window, the amplitude characteristics of the digital filters you applied are shown (characteristics of all the filters you applied are merged).

The information on the filters you have applied is listed in the lower left panel.

You can change the range of the upper figure by pressing the “Change range” button after specifying the upper and lower bounds. If you press this button with a blank textbox. Upper and/or lower bounds are automatically determined.

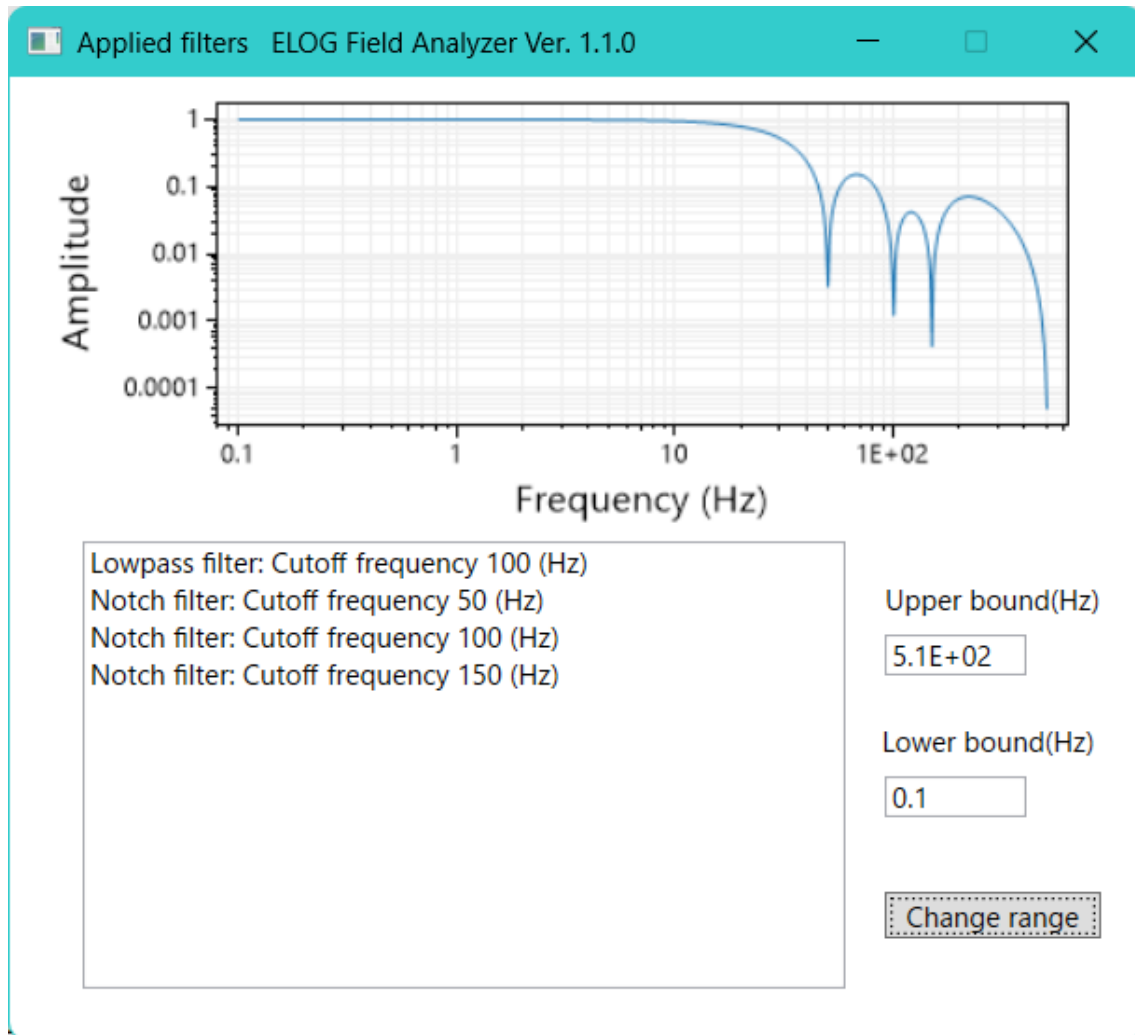


Figure 19: Applied filters window

8.4 Power Spectra Window

When you press the “Power spectra” button, the “Power spectra” window opens. In this window, the power spectra of all channels are displayed.

Only the data of the times displayed in the “Time series” window are used for calculating the power spectra.

The explanation of the items in this window (marked in Figure 20) is written below.

EXPLANATION OF EACH ITEM

- (1) The power spectrum of each channel is displayed. If you double-click the window, the power spectrum will be displayed in a new window, whose size is adjustable.
- (2) The minimum frequency displayed.
- (3) The maximum frequency displayed.

- (4) If you check the box, the linear trend in the logarithmic axis is removed from each power spectrum. If you uncheck, the original power spectra are redisplayed.
- (5) Lower limit of the frequencies to be displayed.
- (6) Upper limit of the frequencies to be displayed.
- (7) The button to change the range of the horizontal axes. If you press the button, the range of the frequency is bounded based on the selection in (5) and (6). If (5) and/or (6) are blank, the range is automatically determined.
- (8) The frequency that the cursor points to is shown here.

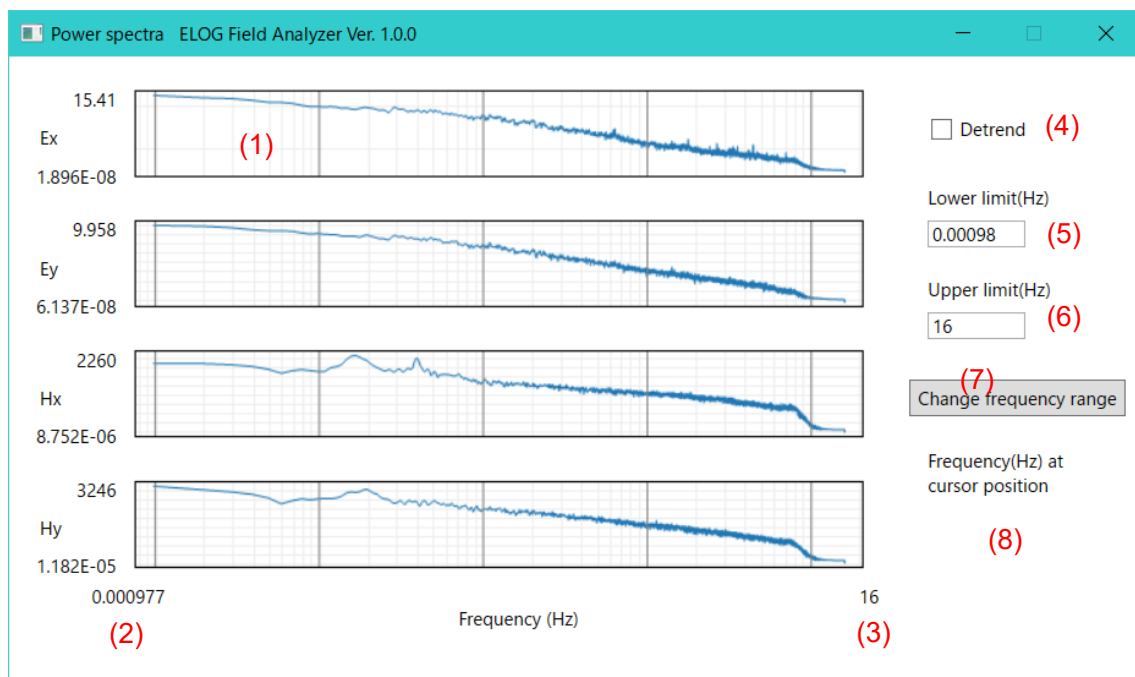


Figure 20: Power spectra window (in the case where the reference magnetic field is not read)

8.5 Response Function Estimation Window

When you press the “Reponses” button, the “Response function estimation” window is opened. In this window, you can select options in the estimation of the response functions, start the estimation, and plot the sounding curves of the estimated response functions.

Only the data of the times displayed in the “Time series” window are used for the response function estimation.

The explanation of the items on this window (marked in Figure 21) is written below.

EXPLANATION OF EACH ITEM

- (1) Work folder. You need to select the work folder for the response function estimation.
- (2) Dipole length for the N-S component of the electric field (unit: meter).
- (3) Dipole length for the E-W component of the electric field (unit: meter).
- (4) Measurement direction from the north (unit: degree). For example, if the +x direction corresponds to the geomagnetic north in the measurement, and the declination around the station is 7 degrees to the west, one needs to write -7 (degrees) in this textbox.
- (5) Rotation angle to the coordinate system in which the response functions are estimated (unit: degree). For example, if you write 40 (degrees) in the textbox, the response functions are estimated in the coordinate system whose +x direction is N40°E.
- (6) Coil calibration file for the Hx component. The file should follow Metronix's calibration file format.
- (7) Coil calibration file for the Hy component. The file should follow Metronix's calibration file format.
- (8) Estimation method. OLS means the ordinary least square. OLS can compute response functions most quickly. If you select “M-estimator” or “RRMS”, outliers are down-weighted. The RRMS estimator is generally the most robust but also the most time-consuming. “RRMS” can be selected only when the remote magnetic field data are read
- (9) If you check it, the ordinary prewhitening method is used, and the response functions are estimated at longer periods than when it is unchecked. This is because the Fourier transforms at the third and fourth frequencies are used if the prewhitening is applied (because the prewhitening can suppress the spectral leakage), whereas the Fourier transforms at the eighth and twelfth fourth frequencies are used when no prewhitening is applied.
- (10) If you check it, the original time series data are downsampled.
- (11) Rate of downsampling. For example, if you wrote 32 for 32 Hz sampling data, the original time series data are downsampled to 1 Hz sampling data.
- (12) Button to make an input file of TRACMT. The input file is created in the work folder you specify in (1). When the input file is successfully made, a message appears at the bottom of the window.
- (13) Button to open the created input file with a text editor. You can check and modify the input file by the text editor.
- (14) Button to start TRACMT to estimate the response function. If you press the button, a command prompt launches, in which the message from TRACMT is displayed
- (15) Button to plot the sounding curves of estimated response functions.

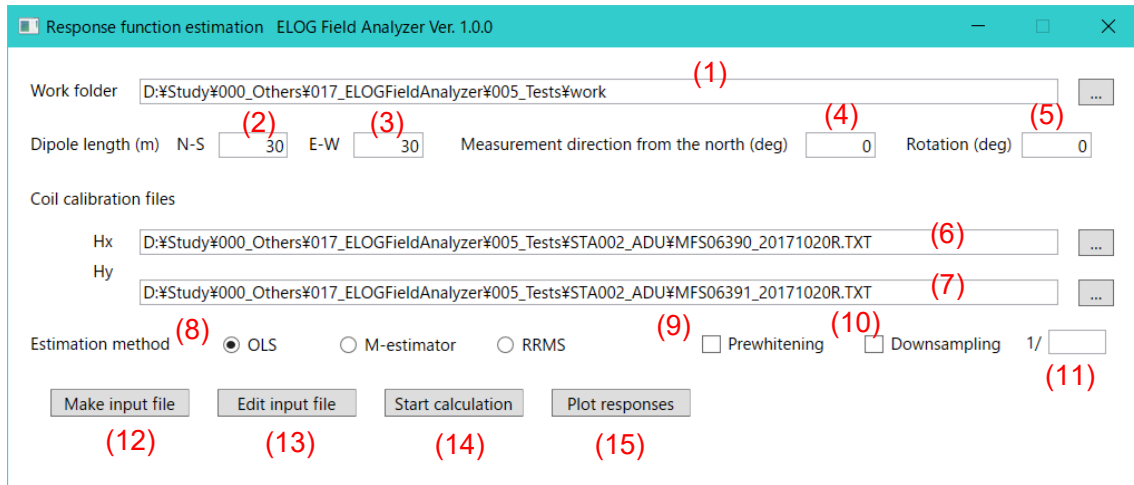


Figure 21: Response function estimation window

8.6 Apparent Resistivity and Phase Window

If you press the “Plot responses” button in the “Response function estimation” window after the calculation of TRACMT is finished, the “Apparent resistivity and phase” window is opened.

In this window, the squared coherence and the sounding curves of the apparent resistivity and phase are depicted

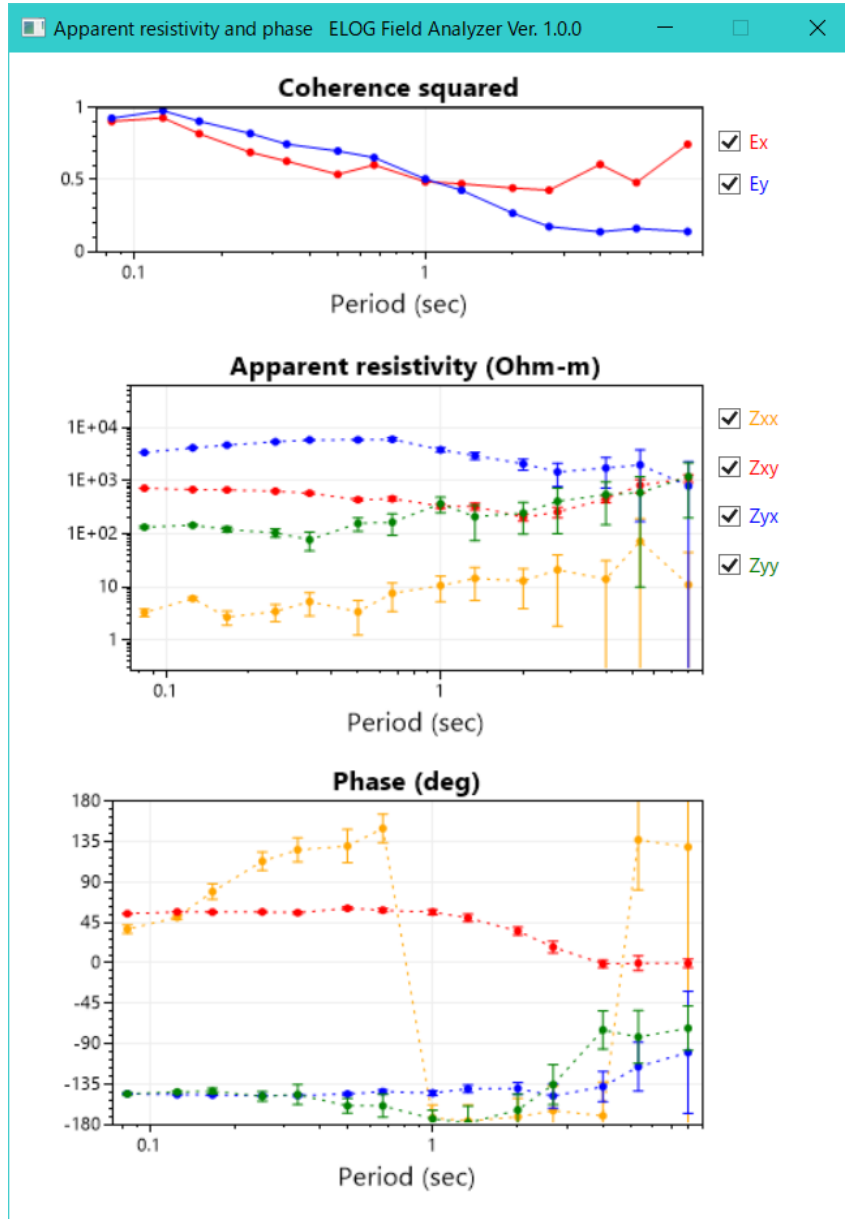


Figure 22: Apparent resistivity and phase window

9. Operation Explanation for ELOG-DUAL (ADU mode) + ATS

9.1 File Selection Window

Please select a folder where dat files of ELOG-DUAL are stored. This folder name must be the measurement day (YYYYMMDD).

You need to select the sampling frequency (32 or 1024 Hz) by the radio buttons on the right side.

If you check the box “Magnetic field data,” the horizontal components (H_x and H_y) of the magnetic field are read from ats files of Metronix instruments. These magnetic field data are used to estimate response functions.

If you check the box “Remote reference”, the magnetic field data is read from ats files of Metronix instruments as remote reference data (H_{rx} and H_{ry}). This reference data will be used in the remote reference method in the response estimation.

CAUTION

In both textboxes for H_x and H_{ry} , you only need to select the ats file for the H_x component. The file for the H_y component is automatically selected in the program. The ats file of the H_x component must contain “_C02_” as the channel index and “_THx_.” Otherwise, the program cannot read the ats file.

The measurement times of all channels should be overlapped at least partly.

Please press the “Read files” button after you properly select sampling frequency and folder and file names.

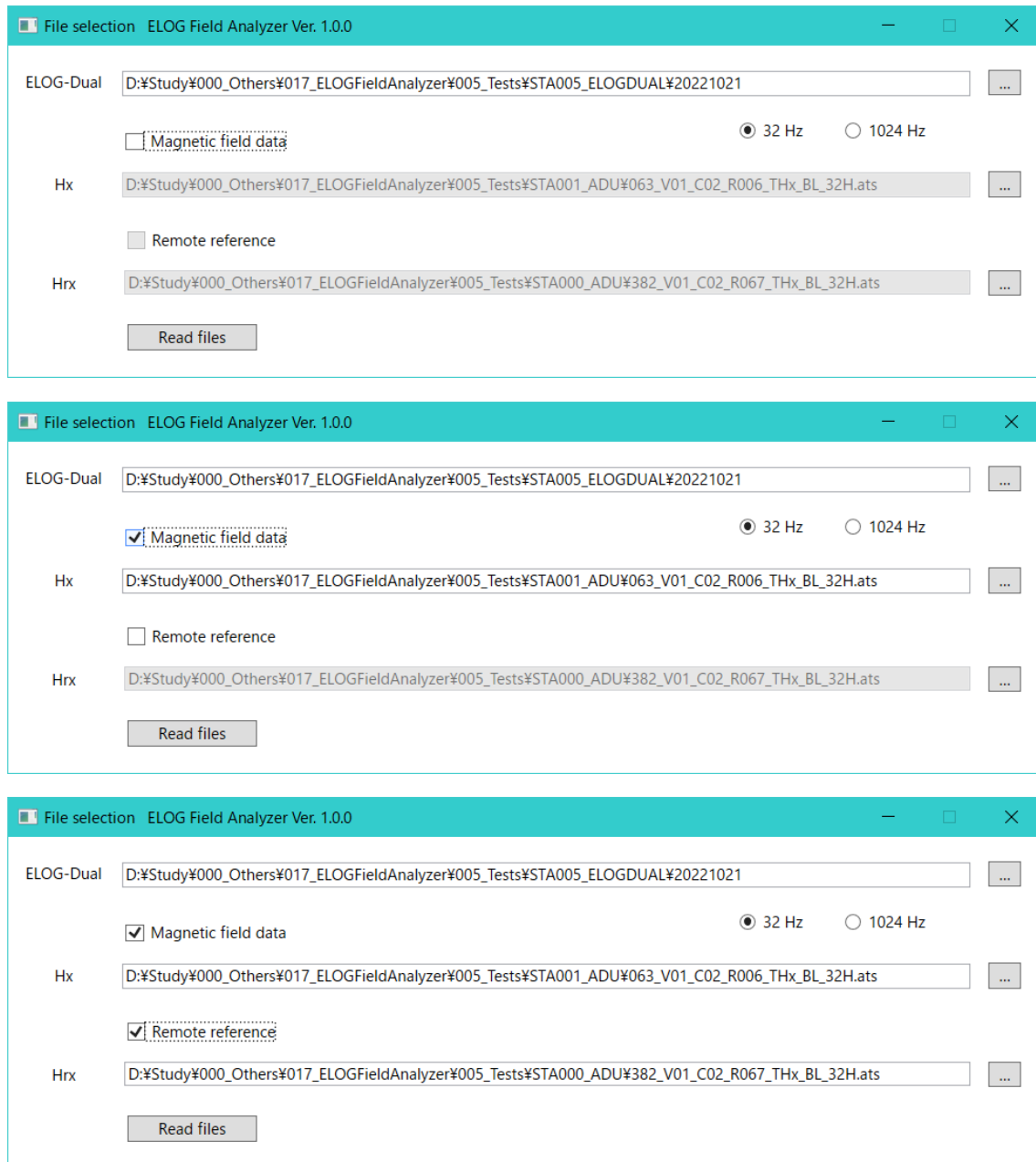


Figure 23: File selection window

9.2 Time Series Window

If dat files are appropriately read, time-series data are displayed.

The explanation of the items in this window (marked in Figure 24) is written below.

EXPLANATION OF EACH ITEM

- (1) The time that the cursor points to is shown in the title bar.

- (2) The time series of each channel is displayed. If you double-click the window, the time series will be displayed in a new window, the size of which is changeable.
- (3) Start time of the time series displayed in the window.
- (4) End time of the time series displayed in the window.
- (5) Button to show the tail end of the time series of the previous day.
- (6) Button to show the time series of the previous day in the same time range.
- (7) Button to show the time series of the next day in the same time range.
- (8) Button to show the head of the time series of the next day.
- (9) Button to decrement the time range. It is not active if the "Type" is "All".
- (10) Slide bar to change the time range. It is not active if the "Type" is "All".
- (11) Button to increment the time range. It is not active if the "Type" is "All".
- (12) Button to apply a low-pass filter.
- (13) Cutoff frequency (Hz) of the low-pass filter.
- (14) Button to apply a high-pass filter.
- (15) Cutoff frequency (Hz) of the high-pass filter.
- (16) Button to apply a notch filter.
- (17) Cutoff frequencies (Hz) of the notch filter. You can write multiple frequencies, comma delimited.
- (18) Clear all digital filters applied.
- (19) Button to show the information on the applied digital filters.
- (20) If you select "All", all of the time series of the specified day are displayed. If "Data number" is selected, the number of data shown in this window is controlled by the selection in (21). If "Data span" is selected, the time span is controlled by the selection in (22).
- (21) The number of data shown in this window. This control is active only if "Type" is "Data number". If you change the number, "Type" is automatically changed to "Data number."
- (22) The time span of the time series shown in this window. This control is active only if "Type" is "Data span". If you change the time span, "Type" is automatically changed to "Data span."
- (23) Button to show the power spectra.
- (24) Button to go to the windows for the response function estimation.

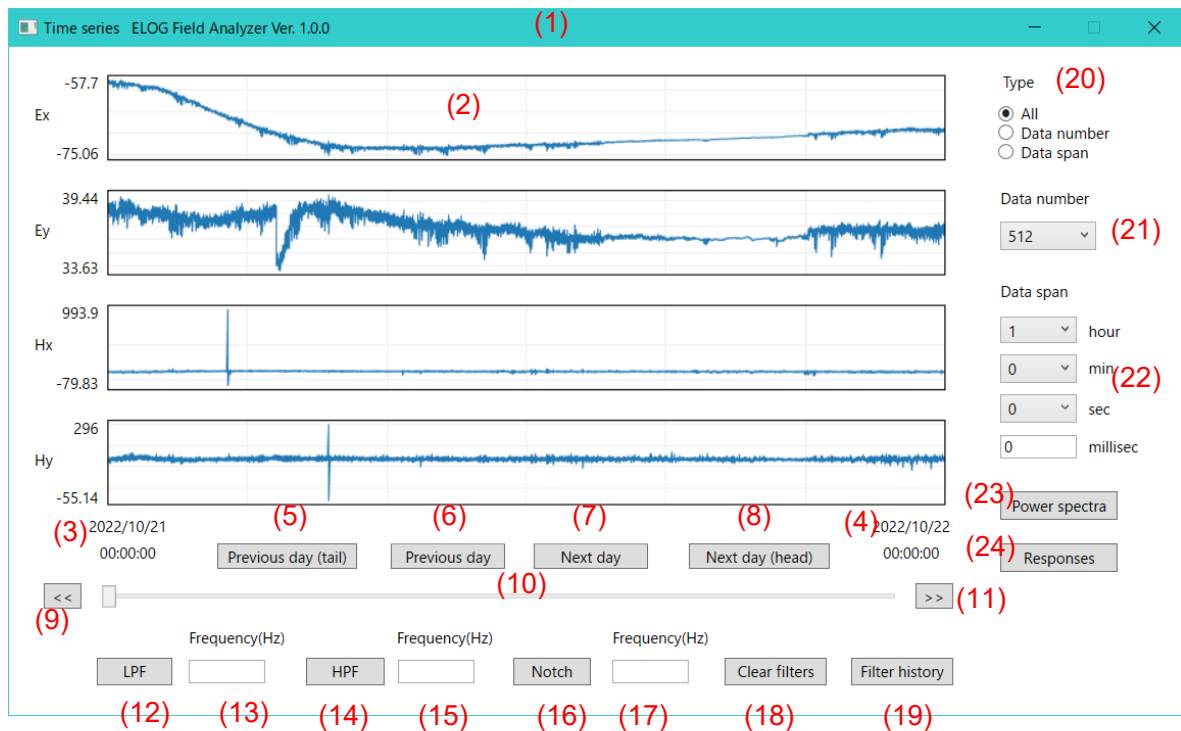


Figure 24: Time series window (in the case where the reference magnetic field is not read)

9.3 Applied Filter Window

When you press the “Filter history” button, the “Applied filter” window will be opened. In this window, the amplitude characteristics of the digital filters you applied are shown (characteristics of all the filters you applied are merged).

The information on the filters you have applied is listed in the lower left panel.

You can change the range of the upper figure by pressing the “Change range” button after specifying the upper and lower bounds. If you press this button with a blank textbox. Upper and/or lower bounds are automatically determined.

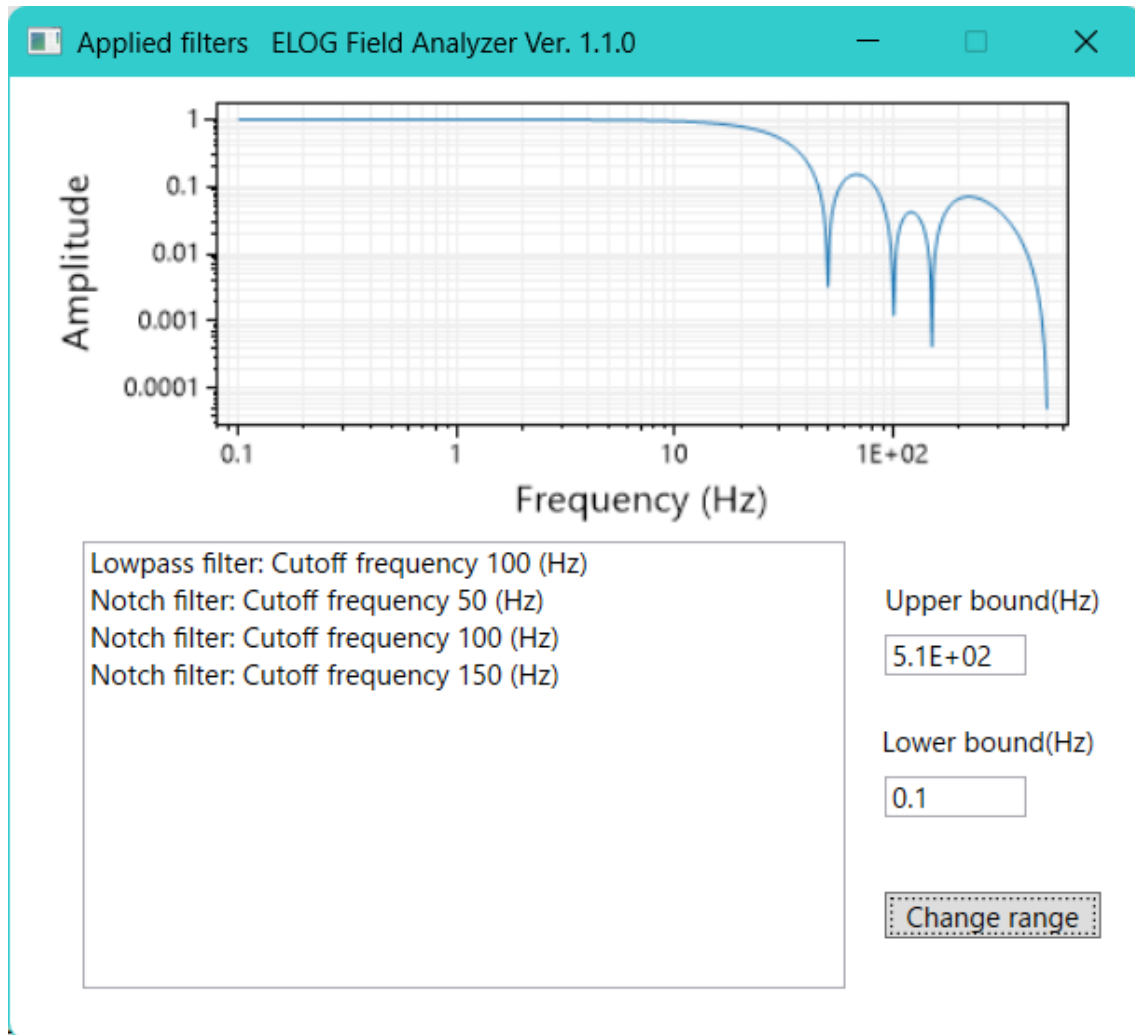


Figure 25: Applied filters window

9.4 Power Spectra Window

When you press the “Power spectra” button, the “Power spectra” window opens. In this window, the power spectra of all channels are displayed.

Only the data of the times displayed in the “Time series” window are used for calculating the power spectra.

The explanation of the items in this window (marked in Figure 26) is written below.

EXPLANATION OF EACH ITEM

- (1) The power spectrum of each channel is displayed. If you double-click the window, the power spectrum will be displayed in a new window, whose size is adjustable.
- (2) The minimum frequency displayed.
- (3) The maximum frequency displayed.

- (4) If you check the box, the linear trend in the logarithmic axis is removed from each power spectrum. If you uncheck, the original power spectra are redisplayed.
- (5) Lower limit of the frequencies to be displayed.
- (6) Upper limit of the frequencies to be displayed.
- (7) The button to change the range of the horizontal axes. If you press the button, the range of the frequency is bounded based on the selection in (5) and (6). If (5) and/or (6) are blank, the range is automatically determined.
- (8) The frequency that the cursor points to is shown here.

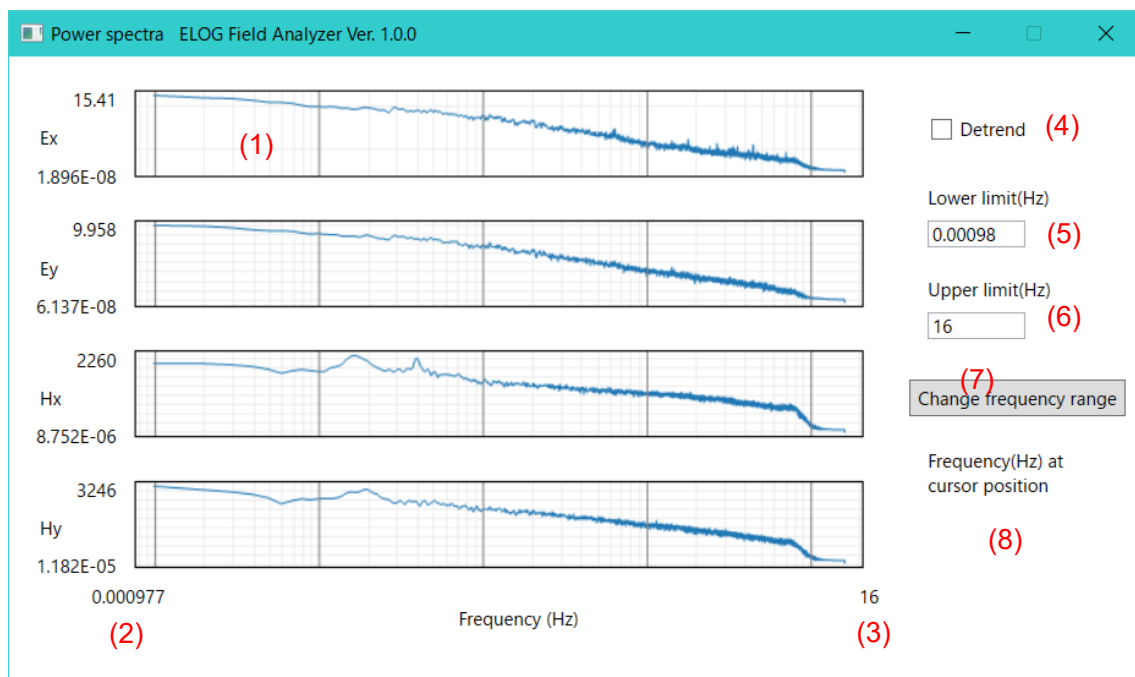


Figure 26: Power spectra window (in the case where the reference magnetic field is not read)

9.5 Response Function Estimation Window

When you press the “Reponses” button, the “Response function estimation” window is opened. In this window, you can select options in the estimation of the response functions, start the estimation, and plot the sounding curves of the estimated response functions.

Only the data of the times displayed in the “Time series” window are used for the response function estimation.

The explanation of the items in this window (marked in Figure 27) is written below.

EXPLANATION OF EACH ITEM

- (1) Work folder. You need to select the work folder for the response function estimation.
- (2) Dipole length for the N-S component of the electric field (unit: meter).
- (3) Dipole length for the E-W component of the electric field (unit: meter).
- (4) Measurement direction from the north (unit: degree). For example, if the +x direction corresponds to the geomagnetic north in the measurement, and the declination around the station is 7 degrees to the west, one needs to write -7 (degrees) in this textbox.
- (5) Rotation angle to the coordinate system in which the response functions are estimated (unit: degree). For example, if you write 40 (degrees) in the textbox, the response functions are estimated in the coordinate system whose +x direction is N40°E.
- (6) Coil calibration file for the Hx component. The file should follow Metronix's calibration file format.
- (7) Coil calibration file for the Hy component. The file should follow Metronix's calibration file format.
- (8) Calibration file for ELOG-DUAL. The format of the calibration file is shown below.
- (9) Folder in which the calibration files for loggers are stored.
- (10) Estimation method. OLS means the ordinary least square method. OLS can compute response functions most quickly. If you select “M-estimator” or “RRMS”, outliers are down-weighted. The RRMS estimator is generally the most robust but also the most time-consuming. “RRMS” can be selected only when the remote magnetic field data are read
- (11) If you check it, the ordinary prewhitening method is used, and the response functions are estimated at longer periods than when it is unchecked. This is because the Fourier transforms at the third and fourth frequencies are used if the prewhitening is applied (because the prewhitening can suppress the spectral leakage), whereas the Fourier transforms at the eighth and twelfth fourth frequencies are used when no prewhitening is applied.
- (12) If you check this, the original time series data are downsampled.
- (13) Rate of downsampling. For example, if you wrote 32 for 32 Hz sampling data, the original time series data are downsampled to 1 Hz sampling data.
- (14) Button to make an input file of TRACMT. The input file is created in the work folder you specify in (1). When the input file is successfully made, a message appears at the bottom of the window.
- (15) Button to open the created input file with a text editor. You can check and modify the input file by the text editor.
- (16) Button to start TRACMT to estimate the response function. If you press the button, a command prompt launches, in which the message from TRACMT is

displayed.

- (17) Button to plot the sounding curves of estimated response functions.

Response function estimation ELOG Field Analyzer Ver. 1.0.0

Work folder: D:\Study\000_Others\017_ELOGFieldAnalyzer\005_Tests\work (1)

Dipole length (m): N-S (2) 30 E-W (3) 30 Measurement direction from the north (deg) (4) 0 Rotation (deg) (5) 0

Coil calibration files

Hx: D:\Study\000_Others\017_ELOGFieldAnalyzer\005_Tests\STA002_ADU\MFS06390_20171020R.TXT (6)

Hy: D:\Study\000_Others\017_ELOGFieldAnalyzer\005_Tests\STA002_ADU\MFS06391_20171020R.TXT (7)

ELOG calibration file

D:\Study\000_Others\017_ELOGFieldAnalyzer\005_Tests\STA005_ELOGDUAL\cal\ELOG_No1.txt (8)

Folder storing other calibration files for loggers

D:\Study\000_Others\017_ELOGFieldAnalyzer\005_Tests\STA005_ELOGDUAL\cal (9)

Estimation method (10) ☒ OLS ☐ M-estimator ☐ RRMS (11) ☐ Prewhitening ☐ Downsampling 1/ (12) (13)

Make input file (14) Edit input file (15) Start calculation (16) Plot responses (17)

Figure 27: Response function estimation window

FILE FORMAT OF ELOG-DUAL CALIBRATION FILE

An example of the ELOG-DUAL calibration file is shown in Figure 28. The first is the header comment line. Below the first line, frequency-dependent characteristics of each channel should be written, comma delimited. The first column is the frequency (Hz). The second and third columns are the amplitudes of Ch#1 (E_x) and Ch#2 (E_y), respectively. The fourth and fifth columns are the phase (deg.) of Ch#1 (E_x) and Ch#2 (E_y), respectively.

	Freq(Hz)	CH1 Amp(Vpp)	CH2 Amp(Vpp)	CH1 Phase(deg)	CH2 Phase(deg)
1	1,4909.68945	4907.04443	0.04834	0.04477	
2	2,4909.55615	4906.88232	-0.44066	-0.44799	
3	3,4909.48779	4906.79687	-0.93957	-0.95059	
4	4,4909.48876	4906.78857	-1.43908	-1.45367	
5	5,4909.45947	4906.72656	-1.9299	-1.94818	
6	6,4909.48291	4906.72167	-2.42168	-2.44362	
7	7,4909.53466	4906.74121	-2.91035	-2.93593	
8	8,4909.55664	4906.73193	-3.41948	-3.44871	
9	9,4909.61816	4906.74804	-3.90501	-3.93786	
10	10,4909.5996	4906.69042	-4.40896	-4.44539	
11	11,4909.6123	4906.65234	-4.89543	-4.93557	
12	12,4909.62207	4906.60009	-5.39715	-5.44087	
13	13,4909.62597	4906.54882	-5.89758	-5.94491	
14	14,4909.62548	4906.46728	-6.39648	-6.44743	
15	15,4909.64501	4906.4248	-6.89529	-6.9498	
16	16,4909.64355	4906.33447	-7.38263	-7.44075	
17	17,4909.60205	4906.2207	-7.88561	-7.94733	
18	18,4909.66503	4906.19775	-8.3786	-8.44391	
19	19,4909.69287	4906.13964	-8.87202	-8.9409	
20	20,4909.45507	4905.88134	-9.37428	-9.44696	
21	21,4909.70312	4905.9497	-9.87922	-9.95512	
22	22,4909.68505	4905.82812	-10.37605	-10.45544	
23	23,4909.71191	4905.74609	-10.87628	-10.95922	
24	24,4909.66992	4905.59472	-11.38133	-11.46779	
25	25,4909.63476	4905.4497	-11.8733	-11.96323	
26	26,4909.61035	4905.28466	-12.37781	-12.47121	
27	27,4909.51708	4905.07031	-12.88148	-12.9783	
28	28,4909.47607	4904.88623	-13.38229	-13.48254	

Figure 28: Example of ELOG-DUAL calibration file

9.6 Apparent Resistivity and Phase Window

If you press the “Plot responses” button in the “Response function estimation” window after the calculation of TRACMT is finished, the “Apparent resistivity and phase” window is opened.

In this window, the squared coherence and the sounding curves of the apparent resistivity and phase are depicted.

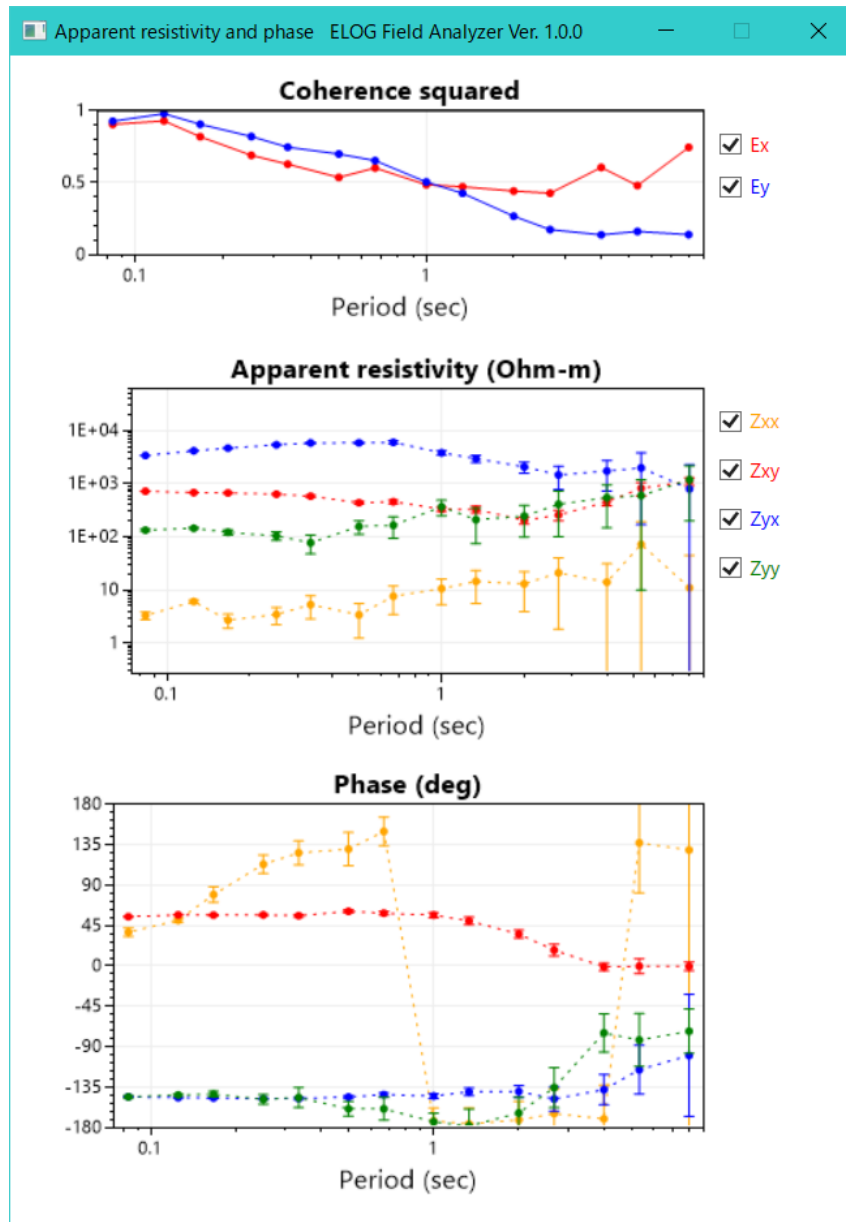


Figure 29: Apparent resistivity and phase window

10. Operation Explanation for ELOG-DUAL (PHX mode)

10.1 File Selection Window

Please select a folder where dat files of ELOG-DUAL are stored. This folder name must be the measurement day (YYYYMMDD).

You need to select the sampling frequency (15, 150, or 2400 Hz) by the radio buttons on the right side.

If you check the box “Magnetic field data,” the horizontal components (H_x and H_y) of the magnetic field are read from dat files of an ELOG-MT station. These magnetic field data are used to estimate response functions.

If you check the box “Remote reference”, remote reference magnetic field data (H_{rx} and H_{ry}) are read from dat files of another ELOG-MT station. This reference data are be used in the remote reference method in the response estimation.

CAUTION

The measurement times of all channels should be overlapped at least partly.

Please press the “Read files” button after you properly select sampling frequency and folder and file names.

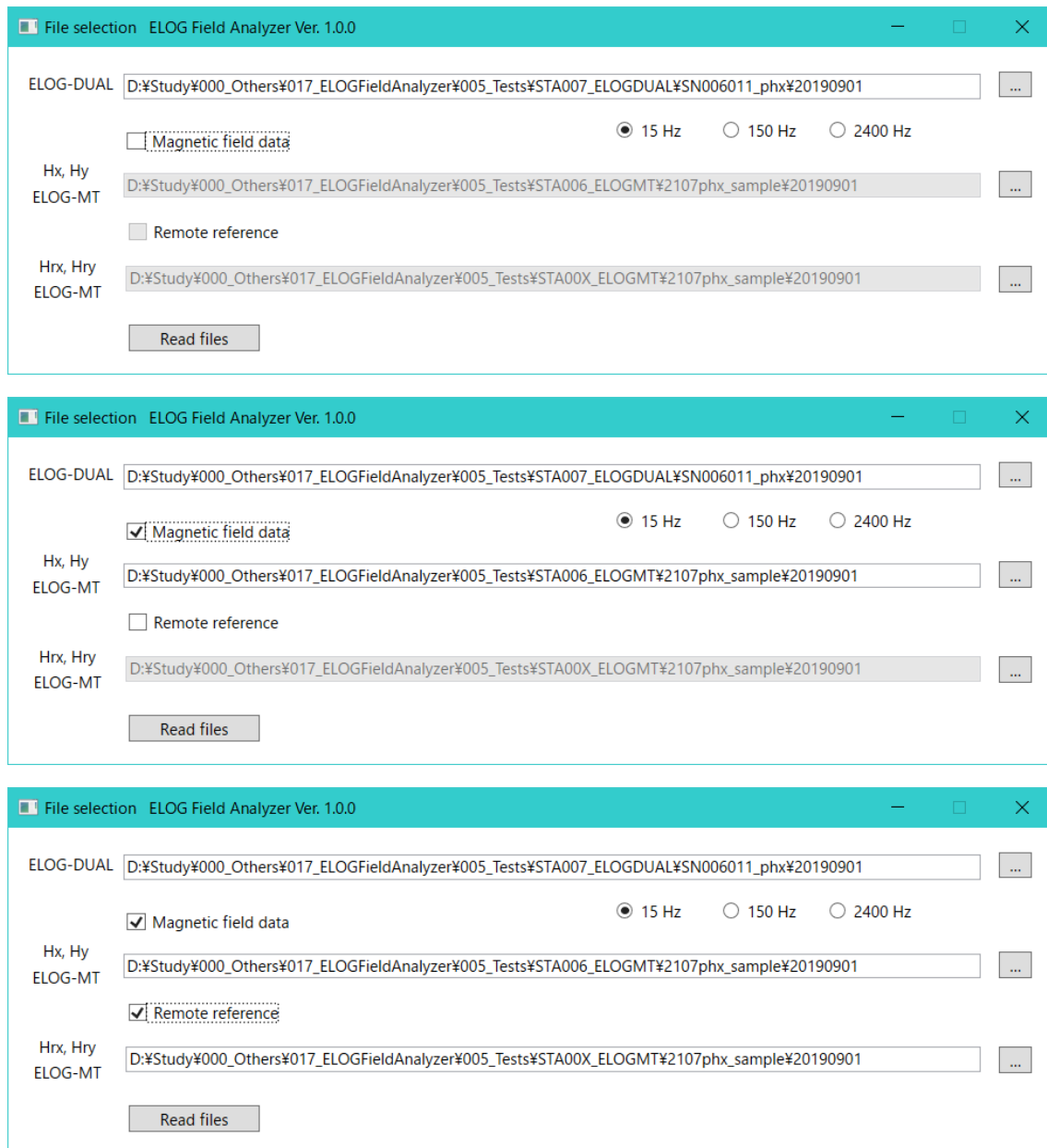


Figure 30: File selection window

10.2 Time Series Window

If dat files are appropriately read, time-series data are displayed.

The explanation of the items in this window (marked in Figure 31) is written below.

EXPLANATION OF EACH ITEM

- (1) The time that the cursor points to the window is shown in the title bar.

- (2) The time series of each channel is displayed. If you double-click the window, the time series will be displayed in a new window, whose size is adjustable.
- (3) Start time of the time series displayed in the window.
- (4) End time of the time series displayed in the window.
- (5) Button to show the tail end of the time series of the previous day.
- (6) Button to show the time series of the previous day in the same time range.
- (7) Button to show the time series of the next day in the same time range.
- (8) Button to show the head of the time series of the next day.
- (9) Button to decrement the time range. It is not active if the “Type” is “All”.
- (10) Slide bar to change the time range. It is not active if the “Type” is “All”.
- (11) Button to increment the time range. It is not active if the “Type” is “All”.
- (12) Button to apply a low-pass filter.
- (13) Cutoff frequency (Hz) of the low-pass filter.
- (14) Button to apply a high-pass filter.
- (15) Cutoff frequency (Hz) of the high-pass filter.
- (16) Button to apply a notch filter.
- (17) Cutoff frequencies (Hz) of the notch filter. You can write multiple frequencies, comma delimited.
- (18) Clear all digital filters applied.
- (19) Button to show the information on the applied digital filters.
- (20) If you select “All”, all of the time series of the specified day are displayed. If “Data number” is selected, the number of data shown in this window is controlled by the selection in (21). If “Data span” is selected, the time span is controlled by the selection in (22).
- (21) The number of data shown in this window. This control is active only if “Type” is “Data number”. If you change the number, “Type” is automatically changed to “Data number.”
- (22) The time span of the time series shown in this window. This control is active only if “Type” is “Data span”. If you change the time span, “Type” is automatically changed to “Data span.”
- (23) Button to show the power spectra.
- (24) Button to go to the windows for the response function estimation.

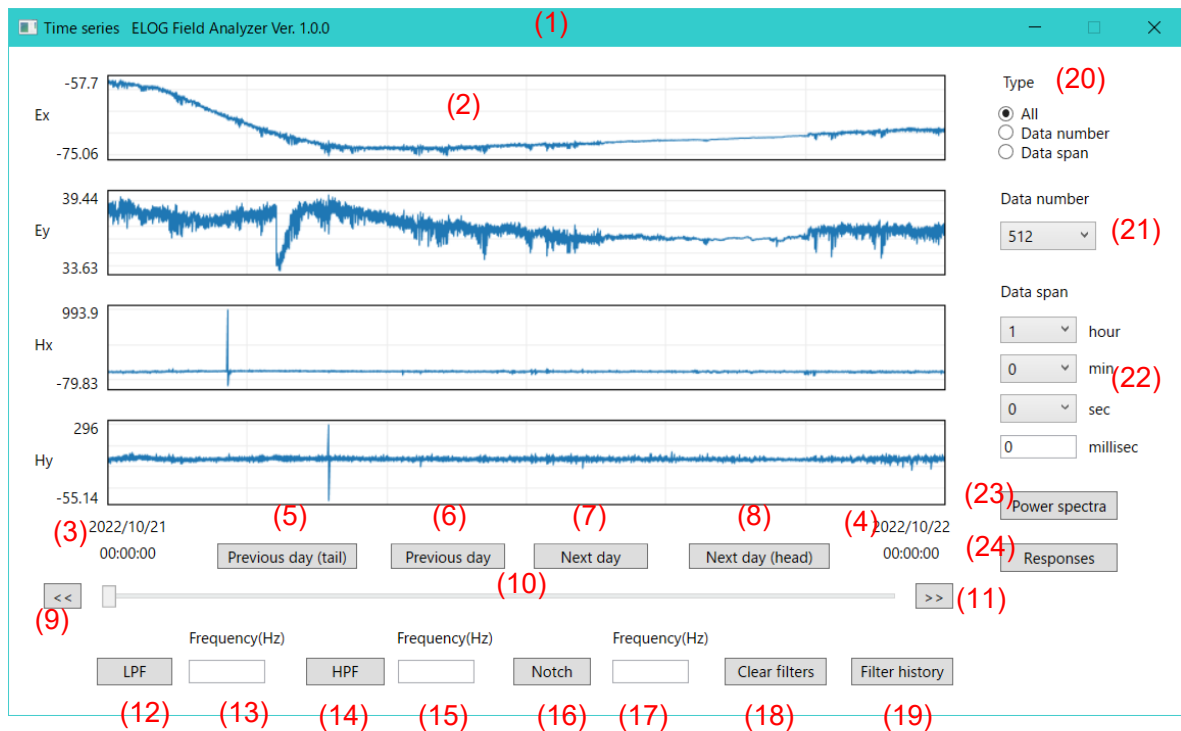


Figure 31: Time series window (in the case where the reference magnetic field is not read)

10.3 Applied Filter Window

When you press the “Filter history” button, the “Applied filter” window will be opened. In this window, the amplitude characteristics of the digital filters you applied are shown (characteristics of all the filters you applied are merged).

The information on the filters you have applied is listed in the lower left panel.

You can change the range of the upper figure by pressing the “Change range” button after specifying the upper and lower bounds. If you press this button with a blank textbox. Upper and/or lower bounds are automatically determined.

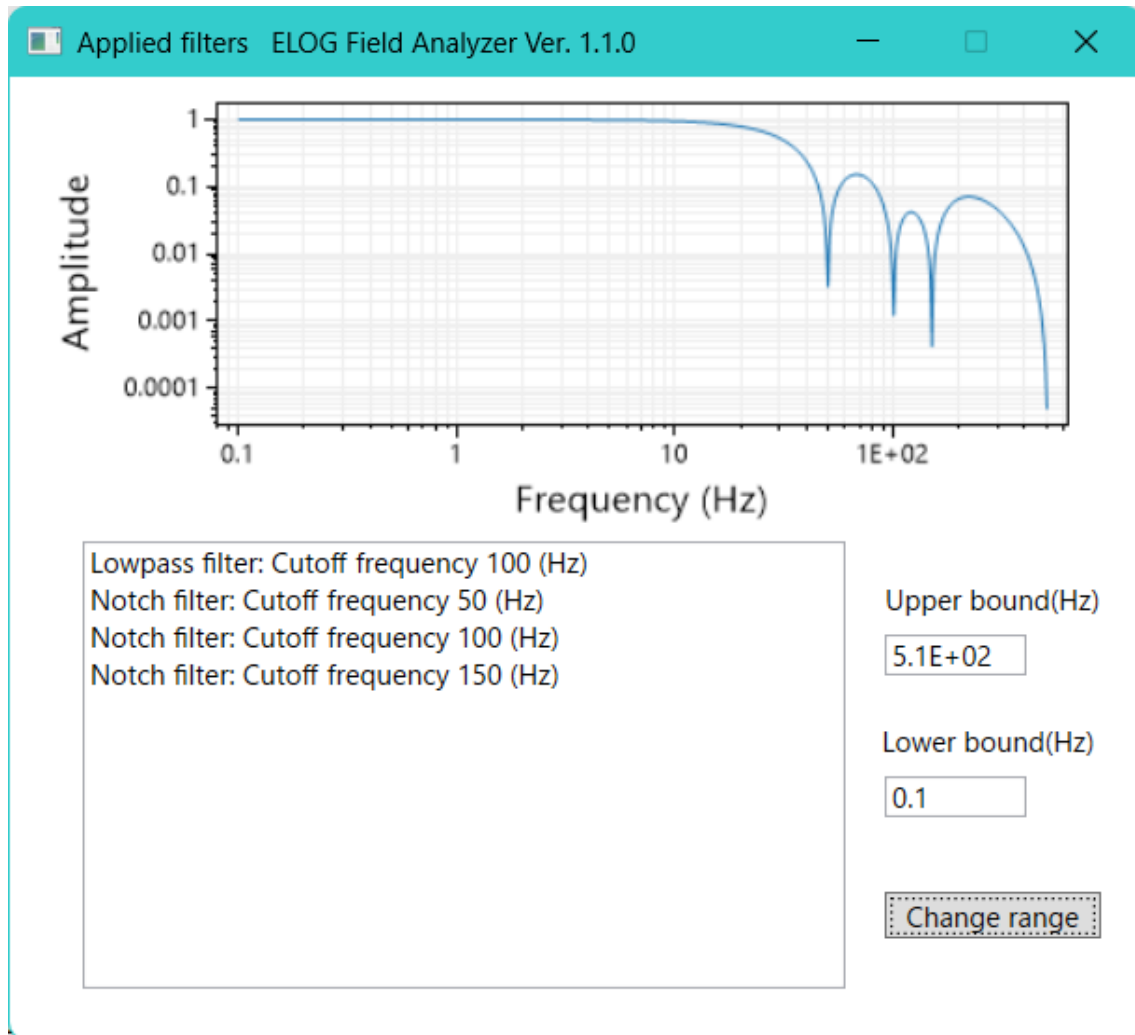


Figure 32: Applied filters window

10.4 Power Spectra Window

When you press the “Power spectra” button, the “Power spectra” window opens. In this window, the power spectra of all channels are displayed.

Only the data of the times displayed in the “Time series” window are used for calculating the power spectra

The explanation of the items in this window, marked in Figure 33, is written below.

EXPLANATION OF EACH ITEM

- (9) The power spectrum of each channel is displayed. If you double-click the window, the power spectrum will be displayed in a new window, whose size is adjustable.
- (10) The minimum frequency displayed.
- (11) The maximum frequency displayed.

- (12) If you check the box, the linear trend in the logarithmic axis is removed from each power spectrum. If you uncheck, the original power spectra are redisplayed.
- (13) Lower limit of the frequencies to be displayed.
- (14) Upper limit of the frequencies to be displayed.
- (15) The button to change the range of the horizontal axes. If you press the button, the range of the frequency is bounded based on the selection in (5) and (6). If (5) and/or (6) are blank, the range is automatically determined.
- (16) The frequency that the cursor points to is shown here.

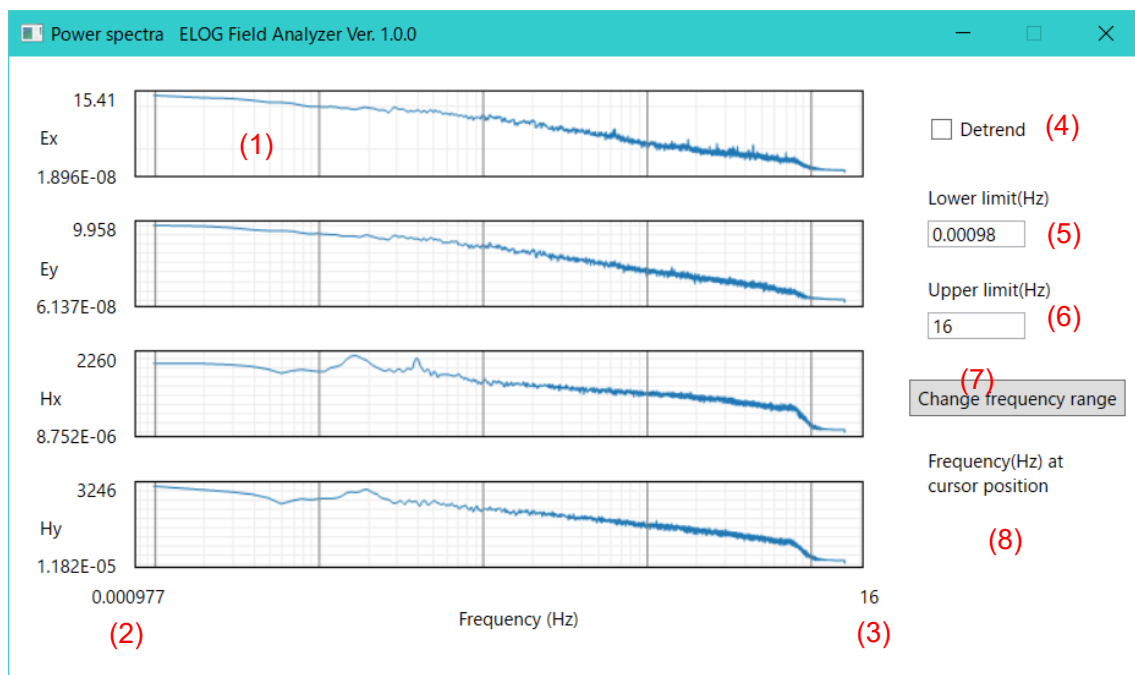


Figure 33: Power spectra window (in the case where the reference magnetic field is not read)

10.5 Response Function Estimation Window

When you press the “Reponses” button, the “Response function estimation” window will be opened. In this window, you can select options in the estimation of the response functions, start the estimation, and plot the sounding curves of the estimated response functions.

Only the data of the times displayed in the “Time series” window are used for the response function estimation

The explanation of the items in this window (marked in Figure 34) is written below.

EXPLANATION OF EACH ITEM

- (1) Work folder. You need to select the work folder for the response function estimation.
- (2) Dipole length for the N-S component of the electric field (unit: meter).
- (3) Dipole length for the E-W component of the electric field (unit: meter).
- (4) Measurement direction from the north (unit: degree). For example, if the +x direction corresponds to the geomagnetic north in the measurement, and the declination around the station is 7 degrees to the west, one needs to write -7 (degrees) in this textbox.
- (5) Rotation angle to the coordinate system in which the response functions are estimated (unit: degree). For example, if you write 40 (degrees) in the textbox, the response functions are estimated in the coordinate system whose +x direction is N40°E.
- (6) Coil calibration file for the Hx component. The file should follow the format described below.
- (7) Coil calibration file for the Hy component. The file should follow the format described below.
- (8) Estimation method. OLS means the ordinary least square method. OLS can compute response functions most quickly. If you select “M-estimator” or “RRMS”, outliers are down-weighted. The RRMS estimator is generally the most robust but also the most time-consuming. “RRMS” can be selected only when the remote magnetic field data are read
- (9) If you check this, the ordinary prewhitening method is used, and the response functions are estimated at longer periods than when it is unchecked. This is because the Fourier transforms at the third and fourth frequencies are used if the prewhitening is applied (because the prewhitening can suppress the spectral leakage), whereas the Fourier transforms at the eighth and twelfth fourth frequencies are used when no prewhitening is applied.
- (10) If you check it, the original time series data are downsampled.
- (11) Rate of downsampling. For example, if you wrote 15 for 15 Hz sampling data, the original time series data are downsampled to 1 Hz sampling data.
- (12) Button to make an input file of TRACMT. The input file is created in the work folder you specify in (1). When the input file is successfully made, a message appears at the bottom of the window.
- (13) Button to open the created input file with a text editor. You can check and modify the input file by a text editor.
- (14) Button to start the response function estimation by TRACMT. After you press the button, a command prompt launches, in which the message from TRACMT is displayed.
- (15) Button to plot the sounding curves of estimated response functions.

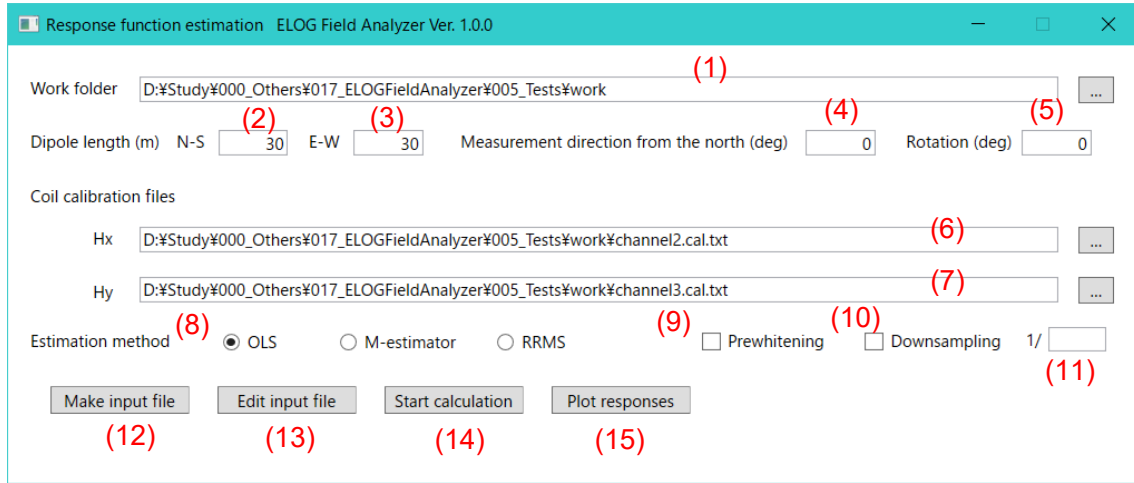


Figure 34: Response function estimation window

FORMAT OF COIL CALIBRATION FILES

Coil calibration files for the magnetic field components should follow the following format.

Instrumental characteristics are corrected in the frequency domain by multiplying A and $C(f)$ as

$$A \times C(f_1)H_x(f_1) \rightarrow H_x(f_1)$$

$$A \times C(f_2)H_x(f_2) \rightarrow H_x(f_2)$$

...

The calibration function is interpolated for the frequency at which $C(f)$ is not specified.

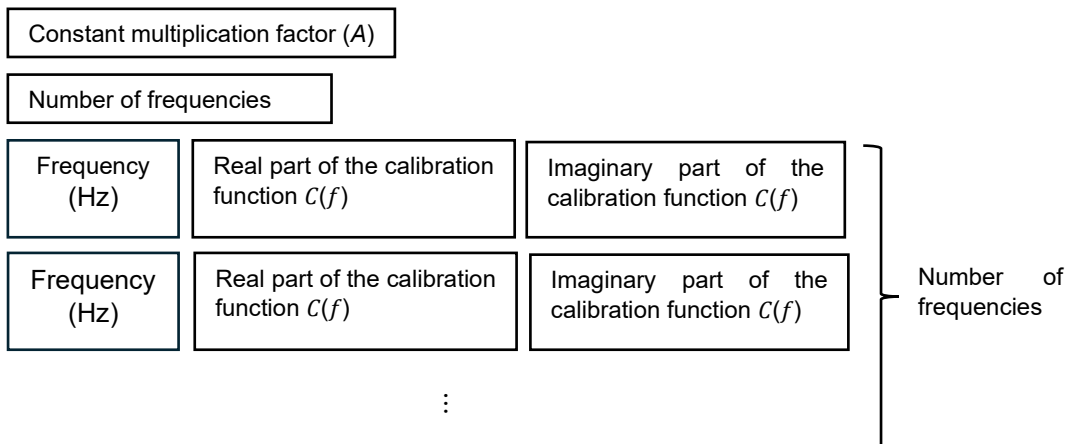


Figure 35: Format of calibration file of TRACMT

10.6 Apparent Resistivity and Phase Window

If you press the “Plot responses” button in the “Response function estimation” window after the calculation of TRACMT is finished, the “Apparent resistivity and phase” window is opened.

In this window, the squared coherence and the sounding curves of the apparent resistivity and phase are depicted

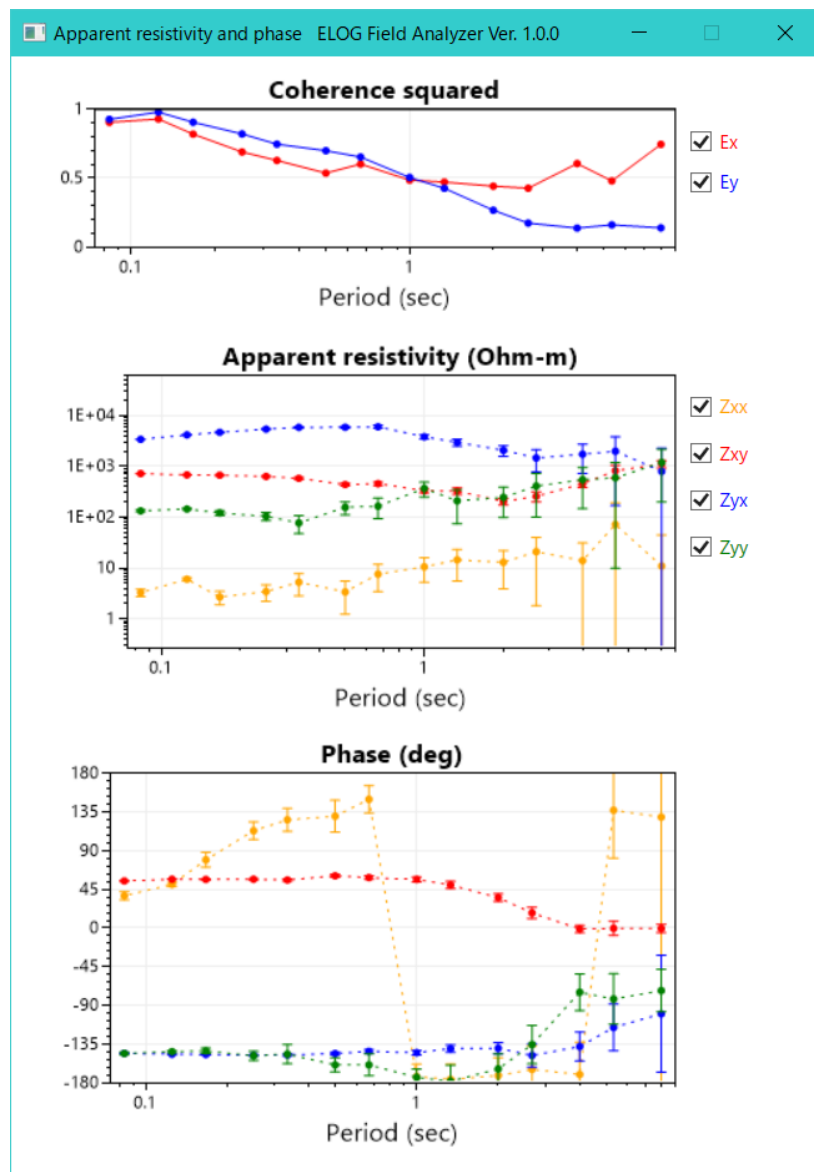


Figure 36: Apparent resistivity and phase window

11. Operation Explanation for ELOG1K + ELOG-MT

Operation for ELOG1K + ELOG-MT is analogous to that for ELOG-DUAL (ADU mode) + ELOG-MT. Please read Section 8.

12. Operation Explanation for ELOG1K + ATS

Operation for ELOG1K + ATS is analogous to that for ELOG-DUAL (ADU mode) + ATS. Please read Section 9.

13. Q & A

Q1: Response functions are not plotted after pushing the “Plot responses” button.

A1: Please confirm that the install and work folders and their parent folders do NOT contain non-ASCII characters (e.g., Japanese characters). If non-ASCII characters are used in those folder names, it is possible that TRACMT does not work properly. The log information from the TRACMT calculation is written to TRACMT.log in the work folder. The log information may provide a clue about the cause of the problem.

Q2: Is it possible to consider differences in the characteristics between channels in estimating response functions?

A2: This program was developed primarily to enable users to check time-series and response functions in the field. To simplify the setting for response function estimation, this program omits the characteristics between channels in estimating response functions, except for ELOG-DUAL (ADU mode) + ATS. However, TRACMT can consider it. To include the characteristics between channels in the calibration, please modify the TRACMT input file (param.dat) as follows, after pushing “Make input file” and before pushing “Start calculation”.

(1) How to consider the characteristics of ELOG-MT

Figure 37 and Figure 38 show the options to consider the characteristics of ELOG-MT and their setting example, respectively. Under the keyword “ELOGMT_CAL”, 0 (ADU mode) or 1 (PHX mode) needs to be written. In the next line, the name of the ELOG-MT calibration file with its path should be written. Figure 39 is an example of the ELOG-MT calibration file. The first line is the comment line. Below the first line, frequency-dependent characteristics of each channel should be written, comma delimited. The first column is the frequency (Hz). The second and third columns are the amplitudes and phase (deg.) of Ch#1 (E_x), respectively. The fourth and fifth columns are the amplitudes and phase (deg.) of Ch#2 (E_y), respectively. In the following columns, the amplitudes and phase (deg.) of the magnetic field channel should be written in the same way. The third line under “ELOGMT_CAL” is a real number used to correct the time difference between loggers. When all data are measured with ELOG, one can write 0.0 because no correction is required.

Under the keyword “LOGGER_CAL_DIRECTORY”, one needs to write the path of the folder in which the filter coefficient files are located. When the PHX mode is used, firh_phx.txt, firm_phx.txt, and fir1_phx.txt should be stored in the folder. When the ADU mode is used, however, firh_adu.txt and fir1_adu.txt should be stored in the folder.

The name of the ELOG-MT calibration file and its path, and the path of the folder in which the filter coefficient files should NOT contain non-ASCII

characters (e.g., Japanese characters).

ELOGMT_CAL
0 (ADU mode) or 1 (PHX mode)
Name of ELOG-MT calibration file with path
Real number for correcting the time difference
LOGGER_CAL_DIRECTORY
Path of the folder in which the filter coefficient files are located

Figure 37: Options to consider the characteristics of ELOG-MT

```
ELOGMT_CAL↓
0↓
../.../Calibration_File/ELOGMT/ELOG-MT_RevB_SN004042.txt↓
0.0↓
LOGGER_CAL_DIRECTORY↓
../.../Calibration_File/Logger↓
```

Figure 38: Example of the setting to consider the characteristics of ELOG-MT

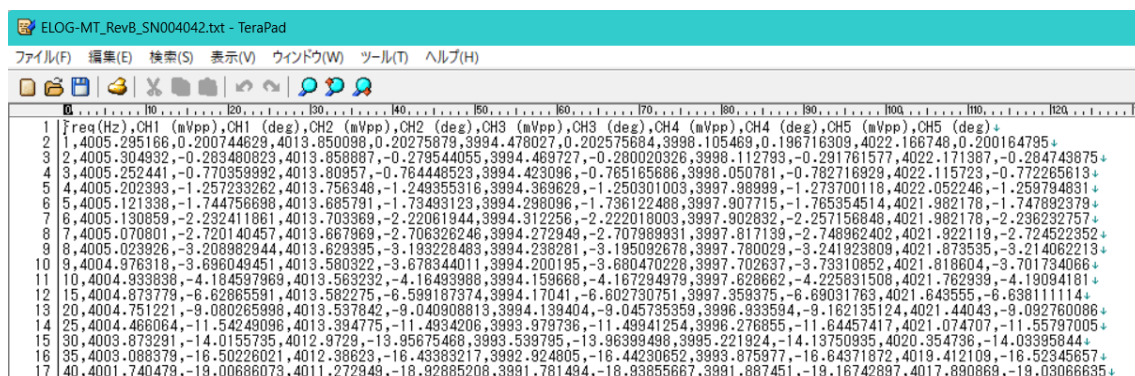


Figure 39: Example of the ELOG-MT calibration file

(2) How to consider the characteristics of ELOG-DUAL/ELOG1K

Figure 40 and Figure 41 show the options to consider the characteristics of ELOG-DUAL/ELOG1K and their setting example, respectively. Under the keyword “ELOGDUAL_CAL”, 0 (ELOG1K), 1 (ELOG-DUAL ADU mode), or 2 (ELOG-DUAL PHX mode) needs to be written. In the next line, the name of

the ELOG-DUAL/ELOG1K calibration file with its path should be written.

Figure 42 is an example of the ELOG-DUAL/ELOG1K calibration file. The first line is the comment line. Below the first line, frequency-dependent characteristics of each channel should be written, comma delimited. The first column is the frequency (Hz). The second and third columns are the amplitudes of Ch#1 (E_x) and Ch#2 (E_y), respectively. The fourth and fifth columns are the phase (deg.) of Ch#1 (E_x) and Ch#2 (E_y), respectively. The third line under “ELOGDUAL_CAL” is a real number used to correct the time difference between loggers. When all data are measured with ELOG, one can write 0.0 because no correction is required.

Under the keyword “LOGGER_CAL_DIRECTORY”, one needs to write the path of the folder in which the filter coefficient files are located. When the ELOG1K is used, firh.txt and fir1.txt should be stored in the folder. When the ELOG-DUAL (ADU mode) is used, however, firh_adu.txt and fir1_adu.txt should be stored in the folder. When the ELOG-DUAL (PHX mode) is used, firh_phx.txt, firm_phx.txt, and fir1_phx.txt should be stored in the folder.

The name of the ELOG-DUAL/ELOG1K calibration file and its path, and the path of the folder in which the filter coefficient files should NOT contain non-ASCII characters (e.g., Japanese characters).

ELOGDUAL_CAL
0 (ELOG1K), 1 (ELOG-DUAL ADU-Mode), or 2 (ELOG-DUAL PHX-Mode)
Name of ELOG-DUAL/ELOG1K calibration file with path
Real number for correcting the time difference
LOGGER_CAL_DIRECTORY
Path of the folder in which the filter coefficient files are located

Figure 40: Options to consider the characteristics of ELOG-DUAL/ELOG1K

```

ELOGDUAL_CAL ↓
0 ↓
../../Calibration_File/ELOG1K/ELOG_No10.txt ↓
0.0 ↓
LOGGER_CAL_DIRECTORY ↓
../../Calibration_File/Logger ↓

```

Figure 41: Example of the setting to consider the characteristics of ELOG-DUAL/ELOG1K

	Freq(Hz)	CH1 Amp(Vpp)	CH2 Amp(Vpp)	CH1 Phase(deg)	CH2 Phase(deg)
1	4909.68945	4907.04443	0.04834	0.04477	
2	4909.55615	4906.88232	-0.44066	-0.44799	
3	4909.48779	4906.79687	-0.93957	-0.95059	
4	4909.48876	4906.78857	-1.43908	-1.45367	
5	4909.45947	4906.72656	-1.9299	-1.94818	
6	4909.48291	4906.72167	-2.42168	-2.44362	
7	4909.53466	4906.74121	-2.91035	-2.93593	
8	4909.55664	4906.73193	-3.41948	-3.44871	
9	4909.61816	4906.74804	-3.90501	-3.93786	

Figure 42: Example of the ELOG-DUAL/ELOG1K calibration file

14. Change History

Ver 1.0.0

Initial release.

Ver 1.1.0

Some figures and explanations are modified. Contents of Q & A are added.